Overview of

Technology and Development of Maglev and Hyperloop Systems

By

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Abstract

Guided transport has higher transport capacity and energy efficiency in operation than other modes of transport, so there has been fast development of guided transport all over the world in the past decades. Although rail transport is the most successful guided transport system, there are still some limitations to the further development of rail transport, e.g. long travel time for long-distance travel, noise and vibration in urban areas, and rail-wheel wear and adhesion. In response, maglev and hyperloop systems which do not rely on the rails and wheels have been proposed and developed.

In maglev and hyperloop systems the vehicles are levitated and propelled from the guideway by magnetic forces. There is no direct physical contact between the moving train and the guideway during operation. To achieve very high-speed operation, hyperloop systems are supposed to run inside low-pressurized tubes to reduce aerodynamic drag. In the world, there are many different types of technology developed to realize maglev and hyperloop systems. However, the technology of maglev and hyperloop systems is relatively new and still under development. Now there are only five low-speed maglev systems and one high-speed maglev system in commercial operation in the world.

The maglev and hyperloop systems have different features compared to the well-developed rail transport, so it is needed to look into their core technology and follow their latest development for future development in the transport sector. In response, this work gives an overview of the maglev and hyperloop systems with respect to their history, technology and applications in the world. This work summarizes the development history of different maglev and hyperloop systems, explain the core technology used in hyperloop systems and three types of maglev systems, and describe the applications of six commercial maglev systems. This work also compares the features of different systems based on statistics. In the end, some conclusions are drawn and future work plans are sketched.

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1 Introduction

Guided transport has often higher transport capacity and energy efficiency in operation than road traffic and air traffic. Since the opening of the first railway service in the UK in 1825, rail transport has been the most important guided transport mode in the world. With rapidly increasing urbanization and high demand for environmental protection from the public, recently the guided transport has given rise to mass transit, in which a large number of people need to be carried. Therefore, there has been fast development of guided transport all over the world in the past decades, especially for high-speed railways and metro systems.

For rail transport, the operational speed covers a very wide range for different travelling purposes, e.g. tramways and metros in urban areas for low-speed operation (30-80 km/h) and high-speed intercity trains for high-speed operation (250-350 km/h). It shows that railway transport is very competitive in short travel distances in urban areas and long-distance travel up to about 800 km over other modes of transport [1].

Travel time is a critical issue for all transport modes. Rail transport has higher running speed than road transport and better accessibility than air traffic. Considering the environmental impact, most railway lines are electrified and sometimes powered by 'green' electricity with zero CO_2 emissions, which makes rail transport more eco-friendly than other modes of transport [2-5]. However, for long-distance travel, the travelling speed of high-speed trains is still too low to compete with air transport on distance longer than ca. 800 km [6-7]. For mass transport in urban areas, noise and vibration are of concern for the people living along the railway lines [8]. These are limitations to the further development of rail transport to compete with other modes of transport.

The rail transport system features rails and wheels. Because of this, the railway system heavily relies on adhesion at the rail-wheel interface to provide guidance, propulsion and braking force. Rails and wheels are relatively stiff and their interaction can result in a series of technical issues related to adhesion, rail-wheel wear, power transmission, noise and vibration, maintenance cost, etc. These issues become more and more significant with increasing speed and are limiting factors to a further increase in operational speed. Even though a French TGV test train proved that a conventional rail-wheel based train can achieve a maximum speed of 574 km/h, the mentioned technical issues still restrict the application of conventional rail-wheel trains for very-high-speed operation in daily commercial operation. In response to these technical limitations some new types of guided transport systems, which do not rely on the rails and wheels, e.g. maglev systems and hyperloop systems, are being developed or have even been used/built, as shown in Figure 1 [9-12].

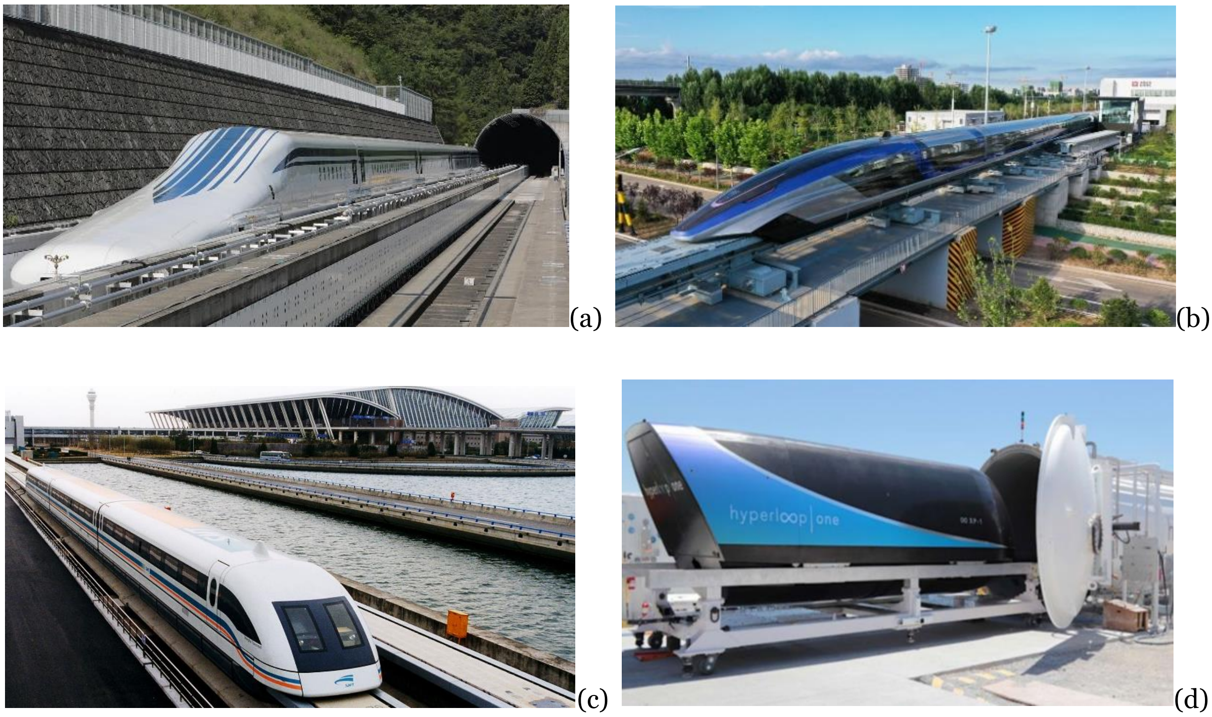


Figure 1: (a) Japanese L0 series maglev train; (b) Chinese HTS maglev train; (c) German Transrapid maglev train operated in Shanghai; (d) US Virgin Hyperloop One.

In order to achieve a very high operational speed, in maglev and hyperloop systems the moving vehicles are levitated and propelled from the guideway by magnetic forces. Since there is no direct physical contact between the moving train and the guideway during operation, the speed is no longer restricted by technical problems linked to the rail-wheel interaction. There are no rotational and sliding components on the moving maglev trains, e.g., wheels, bearings, axles, motors, gears and pantograph, so the rolling resistance and dynamic load are smaller than for conventional trains running at the same speed [13]. Without rotational and sliding components, the corresponding maintenance and reliability, therefore, are improved. Maglev systems are supposed to have higher operational speeds, lower energy consumption, less life cycle costs, less noise and vibration than high-speed trains [14]. However, the aerodynamic drag increases with the square of the speed, so it becomes more and more significant with the increase of the running speed of the train. Although maglev trains have very low rolling resistance and have a streamlined design which can lower the aerodynamic drag, the aerodynamic drag plays the dominating role the very-high-speed operation. In order to overcome the aerodynamic drag at very high-speed operation, hyperloop designs are proposed, where the vehicle/pod with passengers runs inside a large sealed and low-pressurized tube, and boarding and alighting are handled in special terminals [15].

There are many ways to classify maglev systems, e.g. by suspension system, traction system, magnetic system, and operational speed. The common way is to classify the maglev systems according to the operational speed. They can be classified as: low-speed maglev systems (below 100 km/h), medium-speed maglev systems (between 100 km/h and 200 km/h), high-speed maglev systems (between 200 km/h and 600 km/h) and very-high-speed maglev systems (above 600 km/h) [16]. The low-speed and medium-speed systems are designed for mass transport in urban and suburban areas. The high-speed systems are for intercity travels within the medium and long-distance. Since the aerodynamic drag is very high for the very-high-speed operation, the hyperloop system can be considered as an extension of the maglev system for speeds above 600 km/h. In most cases, the hyperloop systems are based on magnetic levitation but operate in a low-air-pressure condition.

Although there are many types of maglev systems developed for low and medium speed, high-speed maglev systems are normally designed to operate at speeds ranging from 400 km/h to 600 km/h. The hyperloop systems are even supposed to operate at speeds above 1000 km/h. Both high-speed maglev and hyperloop systems are supposed to compete with air transport and provide the passengers with both time and energy efficient options in medium and long-distance travelling. The maglev and hyperloop systems have many advantages over the conventional railways and other modes of transport. The infrastructure costs, however, would be much higher than conventional systems. The technology of the maglev and hyperloop systems is relatively new and still under development. Up to now, in the world, there is only one high-speed maglev system in commercial operation, i.e. the Shanghai maglev system with a top operational speed of 430 km/h. There is no full-scale hyperloop line being built. The operational performance of the maglev and hyperloop system is thus not clear when it comes to long-term operation.

This report gives an overview of the maglev and hyperloop systems in the world. The history and development of the maglev and hyperloop systems is described in Chapter 2. Then the theory and key technologies of some key components are summarized in Chapter 3. Application examples and some feedback found in the literature are described in Chapter 4. In the end, some conclusions are drawn and future work plans are sketched in Chapter 5.

2 Development of Maglev and Hyperloop systems

2.1 Conceptual development

The concept of the maglev system started in the early 1900s [17]. High-speed transportation patents were granted to various inventors throughout the world at the beginning of the 20th century. In 1905 United States patents for a linear-motor propelled train were awarded to the German inventor Alfred Zehden. In 1908 Tom Johnson filed a patent for a wheel-less "high-speed railway" levitated by an induced magnetic field. In 1914, the French-born American inventor Emile Bachelet presented his idea of a maglev vehicle and even displayed the first model, as shown in Figure 2.1 [18]. He demonstrated a magnetic suspension using repulsive forces generated by alternating currents, which was the basic idea for electro-dynamic suspension (EDS), but it was not adopted until the superconducting magnets became available because of the high power demand of the electro-dynamic suspension. In 1922 the German researcher Hermann Kemper pioneered an attractive-model maglev system, which was the basic idea of electro-magnetic suspension (EMS). Later, between 1934 and 1941, he was awarded a series of German patents for magnetic levitation trains propelled by a linear motor.

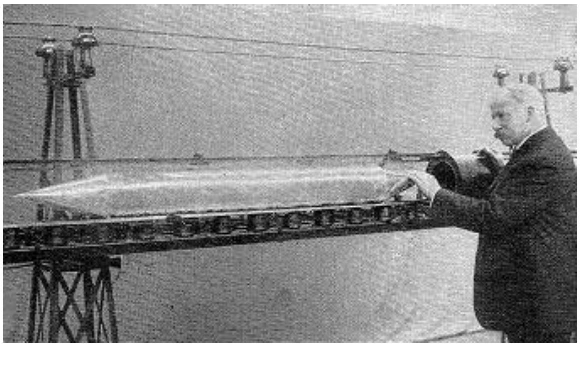


Figure 2.1: Emile Bachelet presenting his maglev model.

Between 1939 and 1943 German engineers worked on a real train based on practical attractive-mode maglev in Göttingen, which finally was presented in 1953. An early maglev train was described in a US Patent "Magnetic system of transportation" by G.R. Polgreen in 1959. The word "maglev" was used in the US patent "Magnetic levitation guidance system" for the first time. After the finding of superconductivity, in 1963 the American researchers Powell and Danby of Brookhaven National Laboratory realized that superconductivity could be used for the maglev system. In 1966 they presented their maglev concept of using superconducting magnets in a vehicle and discrete coils on the guideway, in which the static magnets mounted on a moving vehicle would induce both electro-dynamic lifting and stabilizing forces in specially shaped loops, which realized Bachelet's concept. In 1969, researchers from Stanford developed a continuous-sheet guideway (CSG) concept, in which the moving magnetic fields of the vehicle induced currents in a continuous sheet of conducting material, such as aluminium, to generate a lifting force to the vehicle [19].

Regarding the hyperloop system, tube-based near vacuum transportation was conceptualized over 100 years ago, as a means to achieve very-high-speed ground transport and efficient inter-city travel [15, 20]. An early version was prototyped in 1909 by Boris Weinberg and the concept was published in 1914 in a book, "Motion without friction (airless electric way)". Robert Goddard, a pioneer of rocket design and one of the forefathers of space exploration, proposed a similar concept, which included vacuum pumps to reduce air pressure in a tunnel guideway and a vehicle which was levitated by magnets. With improvements in industrial vacuum pump technology, the vacuum train concept continued development in recent decades. Other proposals included "Evacuated tube transport" (ETT) in the 1990s, "SwissMetro" in the 1990s-2000s, and "Hyperloop" proposed by Elon Musk in 2013. Further developments have been ongoing to make the design practical with added reliability and cost performance.

In 2013 the modern hyperloop concept was proposed by Elon Musk as an innovative transportation method that could be competitive to automobiles and aeroplanes in terms of travel time, speed, and cost for long-distance 'commuting' over distances of 1000 km or more [21]. The concept can be described as a near-sonic train travelling inside an evacuated tube, which attracted a lot of attention from the public as well as from industry and academia.

2.2 Technology development in the world

Since the 1960s, a broader development of maglev systems began in the world. Germany has a long history in the development of maglev systems. Large-scale research and development of maglev transport systems started in 1968 aiming at two different systems, one urban transit system and one intercity high-speed system [22]. Based on Kemper's concept, the first maglev vehicle (Transrapid TR01) was built in 1969 for demonstration, as shown in Figure 2.2(a) [23].

In 1971, the test vehicle (Transrapid TR02) was tested on a 660 m-long test line with a maximum speed of 164 km/h. In 1972, MIT built a 1/25 scale model for testing on a 100 m-long track at speed up to 27 m/s, which is the fundamental of the American Magplane [24]. In the 1960s, along with the development of technologies in other areas, there was much progress on maglev systems achieved in many countries in the world.

Figure 2.2: (a) German Transrapid TR01 maglev train; (a) Japanese ML-100 maglev train; (c) USSR TP-01 maglev train; (d) Birmingham airport international maglev shuttle; (e) Berlin maglev system; (f) Chinese test maglev system in lab.

Following the opening of the Tokaido Shinkansen between Tokyo and Osaka in 1964, the Japanese National Railway started focusing on the development of maglev technology for a higher operational speed [25-26]. In 1972 an experimental vehicle (ML-100) using superconducting technology succeeded in 10 cm-levitation, as shown in Figure 2.2(b). In 1975 the Japanese High-Speed Surface Transport (HSST) finished a prototype maglev train also based on electromagnetic technology and tested it on a 1.6 km-long test track. In the late 1970s, the former USSR began research on maglev systems [27]. In 1979 a 60 m-long test track was built in Ramenskoye, outside Moscow, and a test car (TP-01) on the electro-magnetic suspension was tested on the track, as shown in Figure 2.2(c).

During the 1970s and 1980s, a series of prototype vehicles were built and reached a top speed of 180 km/h. In 1981 the first maglev system for commercial service, the Birmingham airport international maglev shuttle, was built as a shuttle between Birmingham airport and a nearby rail station [28]. In 1984 the system started in operation with a top speed of 36 km/h, as shown in Figure 2.2(d). Today the system is replaced by a cable-propelled double shuttle system, Air-Rail Link. The 1986 World’s Fair (Expo 86) in Vancouver included a short section of a maglev system within the fairgrounds [29]. Germany constructed a maglev system in Berlin (the M-Bahn) that began operation in 1991 to overcome a gap in the city’s public transportation system caused by the Berlin Wall, as shown in Figure 2.2(e) but was dismantled in 1992, shortly after the wall was taken down [30]. In 1993, Korea completed the development of its maglev system with a top speed of 110 km/h, which was shown at the Taejon Expo 93 [31]. In the 1990s, China also started developing maglev systems, as shown in Figure 2.2(f). In 2000 China decided to import the Transrapid maglev system. In 2003 the Shanghai maglev system was opened to the public, which was the first commercialized high-speed maglev system in the world [32].

From 2012 to 2013, a group of engineers from Tesla and SpaceX worked on the conceptual modelling of hyperloop, where the potential design, function, pathway and cost of a hyperloop system were described [15]. In 2017 a 500 m-long test track was completed outside Las Vegas. In 2017 the full-scale passenger pod XP-1 reached a speed of 387 km/h without any passenger in the sealed tube. In 2020 the next generation of passenger pod XP-2 was released and completed the first test with two passengers on board at a speed of 172 km/h. Today there are many countries in the world developing their hyperloop systems.

2.3 German Transrapid systems

Germany has a long history in the research and development of maglev transport systems. The transrapid, M-Bahn and Transurban systems were developed. In 1922 Kemper proposed the concept of maglev transport. In the 1960s, large-scale development of maglev transport systems started for the intercity high-speed transport system, called Transrapid, which was based on electro-magnetic suspension (EMS).

Transrapid is a German maglev technology designed to transport people on high-speed maglev lines. At the late stage, Siemens AG and ThyssenKrupp Transrapid GmbH were behind this project. The development of the Transrapid system was financially supported by the German government. Transrapid is a combination of two Latin words: trans as "over" and rapid which means "fast". The development of the first prototype started in 1969, and the last iteration, the Transrapid 09, was unveiled in 2007 [33]. The Transrapid vehicles are shown in Figure 2.3 [23]. The 32 km-long test track for Transrapid in Emsland was in use since 1980 and closed down in 2011. After decades of work, the Transrapid is no longer in development.

The Transrapid system was developed because maglev operating features are more beneficial for high-speed operation than for low and moderate speed systems. In 1969 an 80 kg vehicle model (TR01) was developed for demonstration. In 1971 the TR02 vehicle was built based on a linear motor with a short primary winding onboard for propulsion. In 1974 the TR03 vehicle was developed with an active control system for guidance. In the same year, the TR04 vehicle was developed by Krauss-Maffei, which was 15 m and 20 t with 20 seats. The TR04 was based on electro-dynamic suspension (EDS) and reached a top speed of 253 km/h in 1977.

In 1977 the TR05 vehicle was built by Thyssen Henschel and Siemens, which was 26 m-long and based on a long primary linear motor in the guideway. The vehicle reached a top speed of 75 km/h and was shown at the Hamburg Expo 1979. In 1983 the TR08 vehicle was developed for an operational speed of 400 km/h, which weighed 120 tons and was 54 meters long. In 1987 a 30 km-long oval test loop was built in Emsland, Germany. The TR06 broke a new Transrapid record in 1988 reaching a speed of 412 km/h, ranked the second-fastest in the world right after the Japanese ML-500R model (517 km/h). In 1989 the TR07 vehicle was developed and reached a top speed of 436 km/h. After a series of inspections and evaluations in 1991, the Transrapid maglev technology was proved to be ready for commercialization.

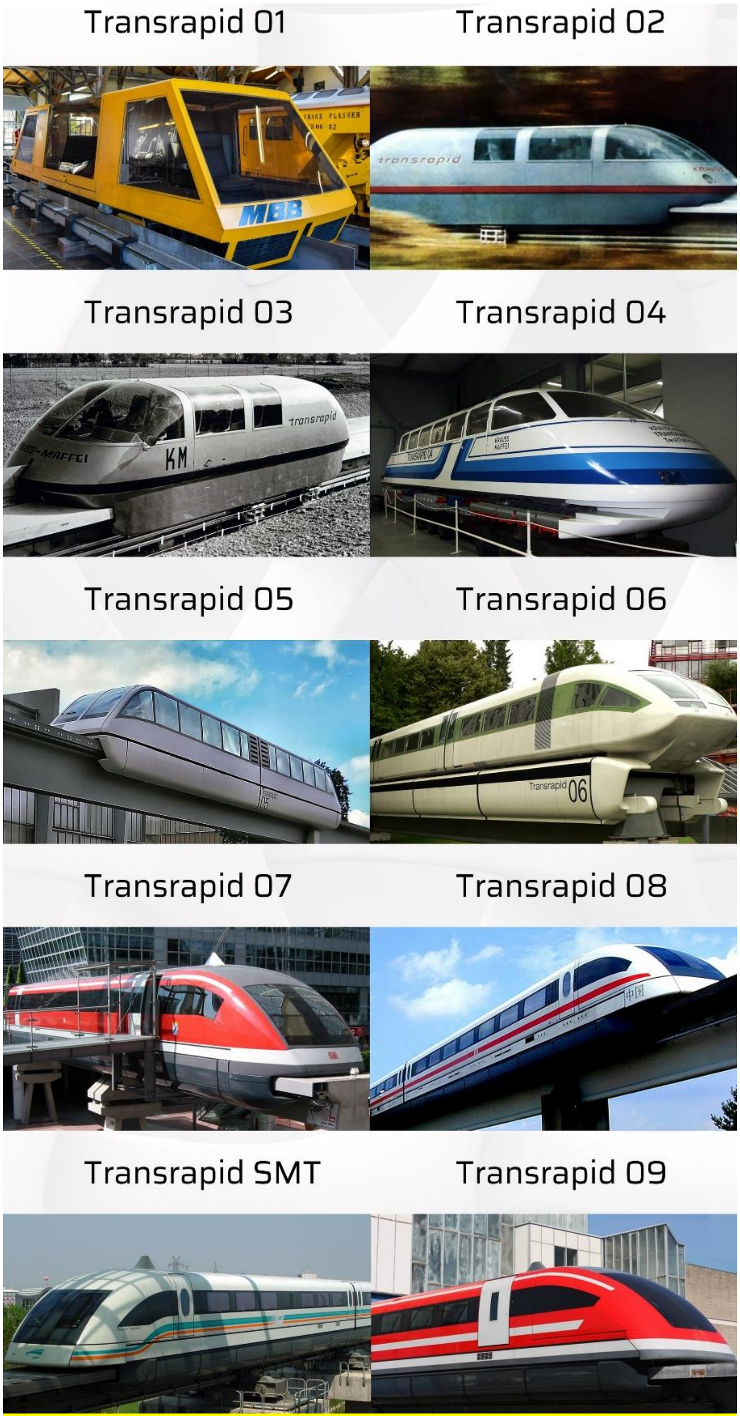


Figure 2.3: Transrapid maglev trains from TR01 to TR09.

In 1992 the Berlin-Hamburg line (292 km-long) was proposed to use the Transrapid maglev system. For commercial operation, the TR08 vehicle was developed with a lighter structure and higher reliability. The three-car train reached a maximum speed of 436 km/h on a test run in 1999 and the test facility was open to visitors for many years. In February 2000, the German government decided to cancel the maglev project due to its low financial viability, and to build the Berlin-Hamburg with high-speed railway technology. In December 2000, Shanghai decided to build the world-first commercial high-speed maglev line. The 30 km-long maglev line started construction in 2001 and was put into commercial operation in 2003. It was operated with a top speed of 430 km/h and in a series of test runs, it reached a top speed of 501 km/h. In 2005 the last Transrapid maglev train TR09 was finished. On 22 September 2006, the Transrapid 08 collided with a maintenance vehicle on the track and the crash killed 22 people. In 2011 the Emsland test track closed down and the last TR09 train was preserved in a museum. The Transrapid maglev system is still the only commercialized high-speed maglev system in the world up to now, which is operated in Shanghai.

2.4 Japanese MLX maglev systems

Japan started its research and development on the maglev system after the opening of the first high-speed railway between Tokyo and Osaka in 1964. The Japanese model used superconductivity and the vehicle-guideway design is based on repulsive magnetic forces, while Transrapid uses attracting magnetic forces. The repulsive suspension technique is inefficient at low speeds so the trains run on rubber tires up to the speed of 100 km/h before becoming magnetically levitated. This dual suspension makes the vehicles more complex, but the tests of high-speed running have proven the technical feasibility of the system. The first prototype vehicle, ML100, was built in 1972. After this, a series of maglev trains have been built and tested, as shown in Figure 2.4 [34].

The first prototype vehicle, ML100, used a short primary linear motor on-board and reached a top speed of 60 km/h. In 1977 a 7 km-long test line with a T-shape structure was built in Miyazaki. The ML500 reached a top speed of 517 km/h on the test line. In 1980 the test track was rebuilt to a U-shape structure for more capacity of the train. The MLU001 maglev train was built and tested and reached a top speed of 400 km/h in 1987. For commercial operation, the MLU002 was developed in 1987 and damaged in a fire in 1991. Then an updated version of MLU002N was built. In 1993 an 18.4 km-long maglev test track was built in Yamanashi for a series of on-track tests. In 1997 the test line was extended to 42.8 km. In 1996 a three-car maglev train MLX01 was developed and reached a top speed of 531 km/h in 1997. In 2009 the maglev system was proved to be ready for commercial operation after a series of technical inspections and evaluations. After nearly two decades of development, the 0-Series was developed for a higher speed in 2002. In 2015 this new maglev train set the world speed record of all trains with the top speed of 603 km/h, which was the highest running speed in the world. The train was built in 2012 by Mitsubishi Heavy Industries and Nippon Sharyo. The operational speed was designed for 500 km/h and the total length was 299 m with a 12-car trainset setup and with a total capacity of 728 seats. In 2014 the Japanese government approved the construction of the 286 km-long maglev line between Tokyo and Nagoya, which is supposed to be finished by 2027. This line will be extended to Osaka in the future.

Figure 2.3: Japanese MLX maglev trains developed from 1972 to 2013.

2.5 Hyperloop

The hyperloop systems concept is that of a vehicle levitating and travelling at high speed and frequency inside low-pressurized tubes, which can help to minimize aerodynamic drag. Pneumatic railways were an early predecessor to this concept with trains affixed with a baffle moving along inside tubes, while fans created a pressure differential in the tubes to deliver force. Functioning versions of this system were installed in London and New York in the 1860s-70s. But these systems pre-dated the large-scale development of vacuum technology and could not take advantage of reduced friction for high-speed inter-city travel.

The latest hyperloop concept was just developed in the last decade thanks to the improvements of industrial vacuum pump technology [15, 35-37]. The hyperloop system was proposed in 2013 by Elon Musk to be competitive to automobiles and aeroplanes in terms of travel time, speed, and cost for long-distance travel. With the advance of levitation systems, high-frequency automation and vacuum infrastructure systems, the technology has grown fast over the last 9 years. Many different hyperloop systems/concepts have been developed in many countries in the world, as shown in three examples in Figure 2.4 [37].

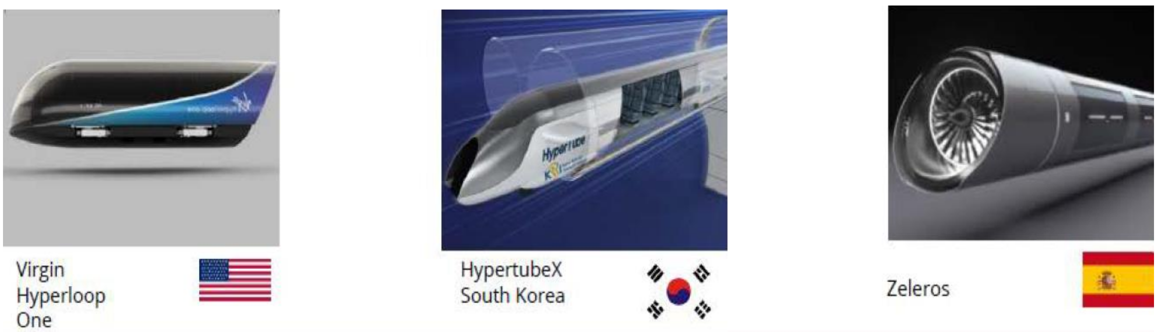


Figure 2.4: Hyperloop systems developed by three different countries in the world.

Since the first modern proposal in 2013, the concept of a hyperloop system has many different variations [15]. For example, small-scale prototypes at the academic/industrial annual competition were sponsored by SpaceX in its specially-built 1.6 km-long track. Large-scale industrial prototypes are expected to be in operation within a decade. The 1.6 km-long test track was built in 2016 in Southern California. Since then the hyperloop pod competition sponsored by SpaceX was held every year, in which many student and non-student teams participate. In 2017 a 500 m-long test track was built outside Las Vegas. The XP-1 test vehicle reached a speed of 387 km/h in the same year [38]. In 2020 the XP-2 test vehicle carried two passengers and reached a top speed of 172 km/h [39]. Since 2018 a 320 m-long full-scale test track started construction in Toulouse, France. Meanwhile, hyperloop systems are developed also in other countries.

3 Theory and Technology

For any kind of guided transport system, there are three basic functions that have to be fulfilled [14]:

- Support,
- Guidance,
- Propulsion and braking.

The support is the vertical contact between the vehicle and the running surface. The guidance relates to the means of steering or lateral motion of the vehicle. The propulsion and braking is the longitudinal force that propels and stops the movement of the vehicle. For conventional rail transport systems, the support, guidance, and propulsion and braking are provided by the rails and wheels. For maglev and hyperloop systems, the vehicle is elevated from the infrastructure and physical contact between the vehicles and the infrastructure is avoided and no rotational components are needed, i.e. the mechanical friction is removed. Therefore, the mechanism of the three basic functions is totally different from the rail transport systems. For the hyperloop systems, to minimize the running resistance caused by aerodynamic drag at high-speed operation, the working environment is also changed due to the tube-based vacuum conditions. No matter what kinds of technical approaches they use, the three aspects are principle requirements to fulfil these functionalities. This chapter will describe the different methods and key components which can realize the functionalities.

3.1 Support

Support is to carry vehicle mass and payload in the vertical direction. For the rail transport systems, the rail vehicles rest their weight through the steel wheels on the rails. For some guided transport systems, rubber tyres are used and stand on concrete/steel beams. However, for the maglev and hyperloop systems magnetic levitation or magnetic suspension is used in which the vehicles are elevated from the guideway by magnetic force to counteract the gravitational force to realize non-contact support.

In magnetic levitation, magnetic materials and systems are able to attract or repel each other with a force dependent on the magnetic field and the area of the magnets. Magnetic levitation is realized through magnetic fields between magnetic objects. Electromagnets, permanent magnets and superconductors can be used to generate the magnetic field in different systems. The maglev and hyperloop systems use these attractive and repulsive forces to lift the vehicle above the guideway. There are several engineering approaches for this in maglev and hyperloop systems.

(1) Electro-magnetic suspension (EMS)

The electromagnet, which is widely used in magnetic levitation systems, is a type of magnet in which the magnetic field is produced by an electric current. Electromagnets usually consist of a large number of closely spaced turns of wire that create the magnetic field. The wire turns are wound around a magnetic core made from ferromagnetic materials. When a current passes through a wire, a magnetic field around that wire is generated. The strength of the generated magnetic field is proportional to the current through the wire. The magnetic core concentrates the magnetic flux and makes a more powerful magnet. The levitation is accomplished based on a magnetic *attractive* force between the guideway and electromagnets as shown in Figure 3.1 [8,14,40]. The attractive force pulls upwards the undercarriage as well as the carbody of the maglev train against gravity. The magnetic rail can be made of permanent magnets or electromagnets. The air gap between the magnetic rail and the electromagnet on the undercarriage is detected by sensors and controlled by adjusting the input current. The electromagnetic attractive force is independent of the running speed, so there could be a constant uplift force even at zero speed. Because the magnitude of the magnetic force decreases much with the air gap increasing, a small air gap around 10 mm is to be maintained.

The EMS system has been widely used in engineering applications, e.g. the Japanese HSST maglev train, the German Transrapid maglev train operating in Shanghai and some low-speed maglev systems in the world. The main advantage of an electromagnet over a permanent magnet is that the magnetic field can be quickly changed by controlling the amount of electric current in the winding. The direction of a magnetic field is dependent on the direction of the electric current. There are also some disadvantages of the EMS. Because the air gap is small between the vehicle and the infrastructure, the structural misalignment in the construction and the structural deformation during operation should be kept as small as possible to maintain the clearance. The electromagnets also consume much electricity to generate the magnetic field.

Once the power supply to the electromagnets is shut down, the levitation stops working immediately. In addition, simply controlling the position usually leads to instability, due to the short-time delays in the inductance of the coil and in sensing the position. In practice, the feedback circuit must use the change of position over time to determine and damp the vertical velocity.

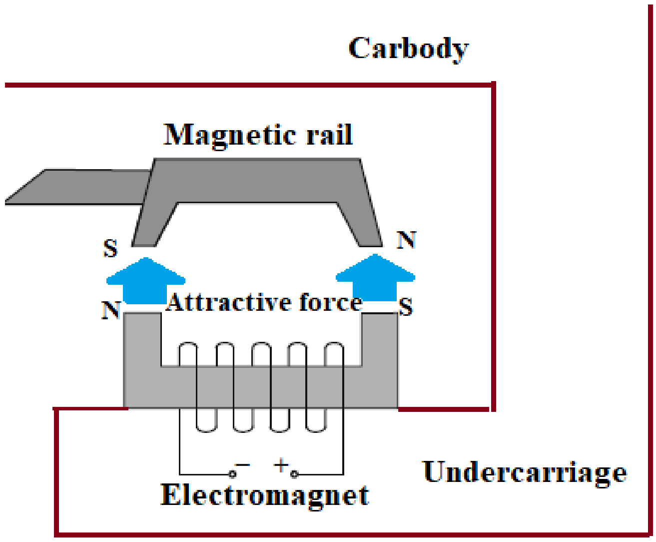


Figure 3.1: Levitation principle of Electro-magnetic suspension (EMS)

(2) Electro-dynamic suspension (EDS)

While the EMS uses an attractive force, the EDS uses a *repulsive* force for levitation. Electrodynamic suspension (EDS) is based on the principle that when the conductors are exposed to time-varying magnetic fields, these fields (relative movement between two objects) can induce eddy currents in the conductors that create a repulsive magnetic field which holds the two objects apart [8,14,17,40]. When the magnets attached to the trains move forward on the inducing coils or conducting sheets located on the guideway, the induced currents flow through the coils or sheets and generate the magnetic field, as shown in Figure 3.2. The trains are elevated by the repulsive force between the magnetic field on the guideway and the magnets on the train. The magnetic field on the train is produced by either permanent magnets or superconducting magnets. The induced magnetic field is generated by wires or other conducting strips on the track. The magnets and the induced conductors do not need to be mounted below the carbody but can be mounted on the two sides below the centre of gravity of the carbody, see e.g. the Japanese MLX maglev system.

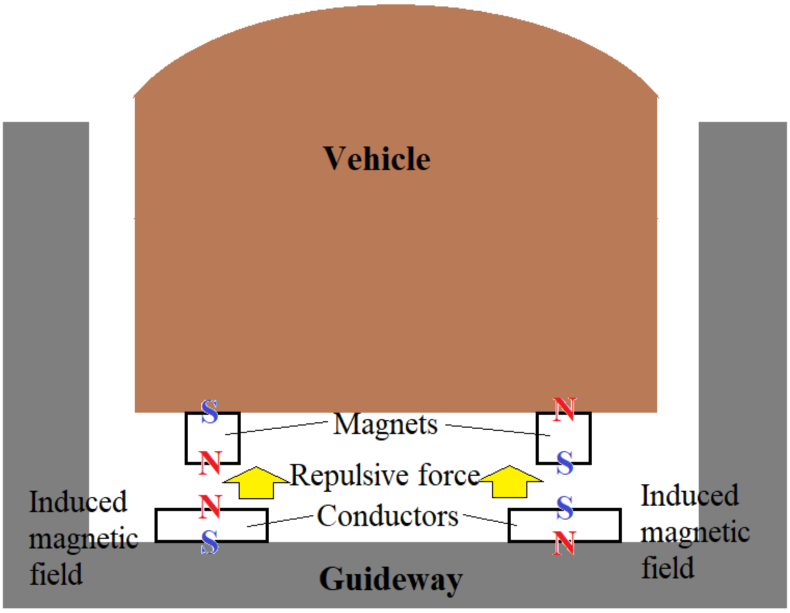


Figure 3.2: Levitation principle of Electro-dynamic suspension (EDS).

At low speeds, the current induced in the conductors is not large enough to produce a sufficient repulsive electromagnetic force to support the weight of the train due to the slow change in magnetic flux with respect to time. Moreover, the energy efficiency for EDS at low speed is low. For this reason, the train must have wheels or some other form of take-off and landing gear to support the train until it reaches a speed (around 100 km/h) that can sustain levitation. Since a train may stop at any location, due to equipment problems, for instance, the entire track must be able to support both low-speed and high-speed operation. At high speeds, the induced repulsive force becomes vigorous and stable so that the air gap is increased to around 100 mm and reliable for the variation of the load, so it is unnecessary to control the air gap and magnetic field. The air-gap can cope with small and medium structural irregularities or structural deformations. Therefore, EDS is highly suitable for high-speed operation.

By the magnets on the trains, the EDS may be divided into two types: the permanent magnet (PM) type and the superconducting magnet (SCM) type. For the PM type, the system is relatively simple because there is no need for a high electric power supply. But the PM type is only limited to applications for small systems because there are no high-powered PMs available. For the SCM type, the system is relatively complex to maintain a very low temperature for superconduction by evaporation of liquidized helium. Even though the power supply to the vehicle is turned off, the super-cooled coils can maintain the levitation for a while, which is different from the EMS. The generated heat of the induced currents may cause problems during operation. The Japanese MLX maglev system used the EDS design for levitation and achieved a maximum running speed of 603 km/h in 2015.

(3) Superconducting magnetic suspension (SMS)

EMS and EDS have been developed and partly put into commercial operation for many years. In order to elevate the vehicle from the guideway, a complex suspension system has to be onboard that consumes a great amount of energy. For example, for EMS, electric energy is continuously needed to magnetize the electromagnets and a control system is used to maintain the air gap within an acceptable range, while EDS does not work at low speeds and needs a complex and expensive cooling system for liquid helium for superconductors (Japanese MLX maglev system). But with the development of high-temperature superconductors, high-temperature ceramic superconductors that can be cooled by using liquid nitrogen rather than liquid helium for superconductors, it is possible to use high-temperature superconductors and permanent magnets to achieve levitation with an inexpensive coolant [41-43]. The levitation does not require a control system to maintain the air gap and is not dependent on running speed (i.e. working at the entire speed range).

According to a design proposal, a permanent magnet guideway (PMG) is used to provide the applied magnetic field and the high-temperature superconductors are mounted below the vehicle. The repulsive force between the superconductors and the permanent magnets can lift up the vehicle from the guideway at any speed (even when the vehicle is standing still), as shown in Figure 3.3. There is neither energy nor air-gap control system needed for levitation. The air gap can be maintained at around 10-20 mm (for the Chinese HTSS prototype maglev train).

The HTSS system is under development and has not yet been commercialized up to now. The advantage of the HTSS system is self-stabilizing levitation without additional energy consumption. The HTSS system onboard is lighter and simpler than the EMS and EDS systems.

The HTSS system still has some limitations. First of all, the loading capacity is still small (supportive force density of superconductor: ca. 6 N/cm^2), and the small air gap requires small structural misalignments in the construction and low deformation during operation. Secondly, the permanent magnet guideway uses a great number of permanent magnets which is made from rare-earth elements and leads to a high capital cost of the infrastructure. Lastly, the permanent magnets have a strong attraction to ferromagnetic metals, so only low-magnetic steel can be used for the infrastructure. It is very important for a safe and long-term operation to keep the guideway clear from all kinds of ferromagnetic metal.

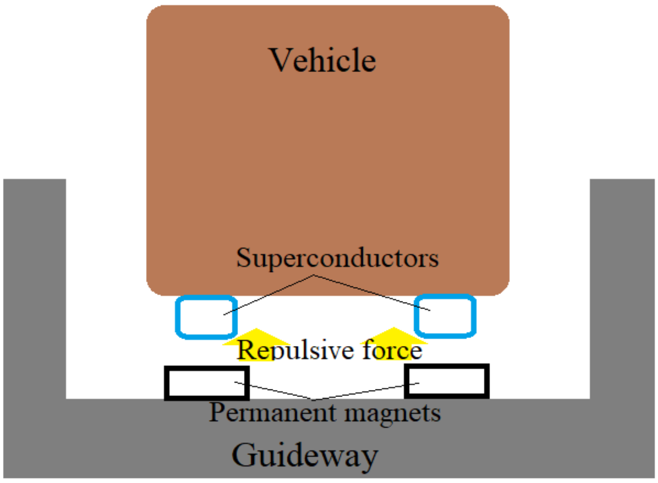


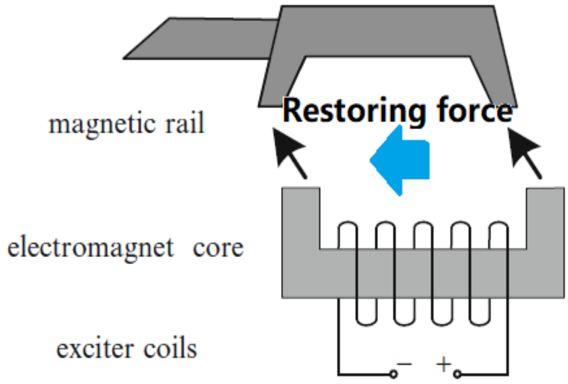
Figure 3.3: Levitation principle of high-temperature superconducting suspension (HTSS).

3.2 Guidance

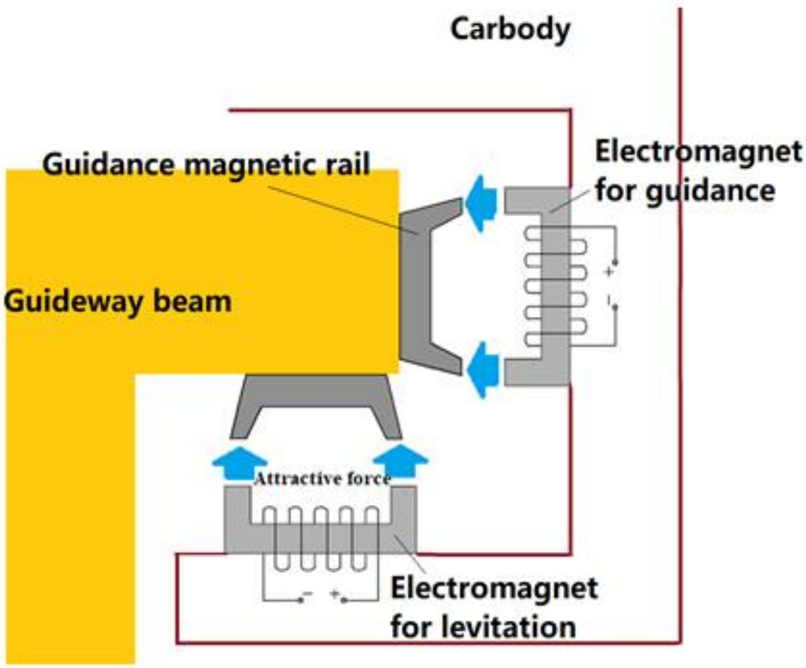
Guidance means the lateral steering of the vehicle. For guided transport systems, the travelling course and the lateral movement of the vehicle are defined and constrained by the guideway. This distinctive feature makes the guided transport systems much different from other modes of transport, e.g. cars, aeroplanes and vessels. For railway systems, the conicity of the wheels ensures that the solid-axle wheelsets are self-steering and that the wheel flanges only provide a backup constraint in the lateral direction. Since maglev and hyperloop systems should also be guided by the guideway, a reliable guidance mechanism is very important. For maglev and hyperloop systems, the physical contact between vehicles and guideway is avoided, so a magnetic force is also used for guidance to counteract the centrifugal force in curves and withstand any possible disturbances in the lateral direction. There are three guidance approaches corresponding to the three types of suspension above.

(1) Electromagnetic suspension (EMS)

For low-speed maglev systems, both the operational speed is low and the centrifugal force in curves is small. The EMS is self-centred, which means any lateral displacement between the magnetic rail and the electromagnet leads to a restoring force to pull the vehicle back to the centre position, as shown in Figure 3.4(a) [8,14]. The magnetic field can be adjusted to strengthen the restoring force by detecting the air gap. Therefore, there is no need to have a specific guidance system onboard or in the infrastructure.



(a)



(b)

Figure 3.4: (a) Restoring force of low-speed EMS maglev system for guidance; (b) Additional lateral suspension for high-speed EMS maglev system.

However, for high-speed maglev system, the operational speed, centrifugal force and dynamic load are very high. In addition, the air gap between the vehicle and the infrastructure for the EMS system is small, so the restoring force is not strong and fast enough to centre the vehicle back to its original position. In order to acquire a high restoring force, another electromagnetic suspension is used in the lateral direction, as shown in Figure 3.4(b). The system works as the EMS for levitation. There are also sensors to detect the air gap in the lateral direction to control the electric current through the electromagnets.

(2) Electro-dynamic suspension (EDS)

As mentioned, the electro-dynamic suspension uses repulsive force for levitation. The repulsive force is very strong with speed increase and can maintain a large air gap (clearance) between the infrastructure and the vehicle, so it is not necessary to place the magnets and the induced conductors right below the carbody. In order to provide the restoring force in the lateral direction for guidance, the EDS is placed on the two sides of the vehicle and the induced conductors are a bit below the magnets onboard, as shown in Figure 3.5 [8,14,44]. In that case, the same suspension system works for both vertical levitation and lateral guidance.

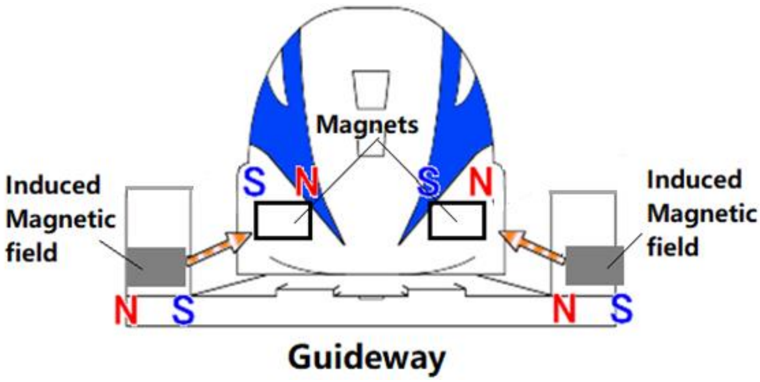


Figure 3.5: Electrodynamic suspension for guidance.

Even though the suspension system becomes simple, as mentioned previously, the EDS does not work well in the low-speed range, so wheels or a special gear are needed not only right below the carbody but also on the two sides of the vehicle. In an emergency condition, they have to take over the EDS system to carry the train weight. The suspension systems, therefore, become relatively complex.

(3) Superconducting magnetic suspension (SMS)

For the superconducting magnetic suspension (SMS), the superconductors are mounted onboard and the permanent magnets are used to build the guideway [41-42]. When the superconductors are exposed to a magnetic field, the so-called Meissner effect comes into play, which means that the superconductors oppose any change to the externally applied magnetic field [45]. For the SMS system, any lateral movement of the superconductors results in a strong restoring force to withstand the lateral movement, because the magnetic flux of the permanent magnets becomes small when the distance between the superconductors and the permanent magnets is increasing apart. Therefore, the lateral movement of the vehicle is constrained. Since the guideway is built with identical permanent magnets in the longitudinal direction, the magnetic flux along the guideway is not changed, so the vehicle can move freely in the longitudinal direction, as shown in Figure 3.6.

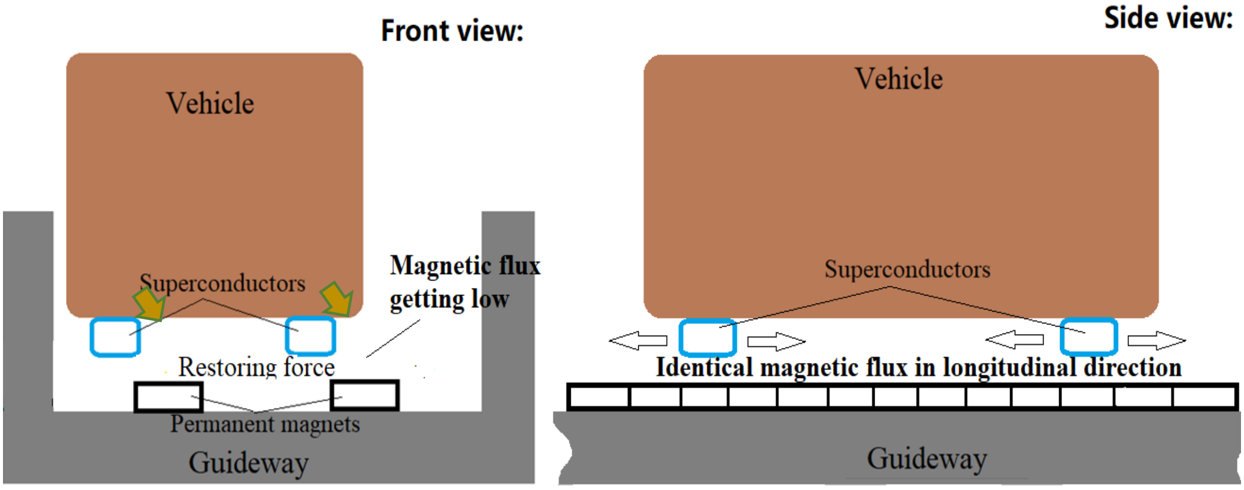


Figure 3.6: Superconducting magnetic suspension (SMS) for guidance.

The effect is a welcomed feature of the SMS, which can directly be used for guidance without adding another guidance system for the maglev trains. However, the restoring force is constant and does not increase with speed, so the restoring force of the superconductors is not high enough for high-speed operation. Further studies on SMS are needed before it can be used for commercial applications.

3.3 Propulsion and braking

The propulsion is the longitudinal force that propels the vehicle. For other kinds of guided transport systems, the propulsion force is generated at the contact interface between the moving vehicle and the guideway, e.g. the rail-wheel interface for the railway systems and the tyre-guideway interface for the rubber-tyre trains. The physical contact between the rotating wheels and the guideway results in rolling resistance and wear. The adhesion available for the propulsion is not constant but subjected to many influencing factors, e.g. decreased adhesion at high speeds and at contaminated surfaces. For maglev and hyperloop systems, the physical contact between the vehicles and the guideway is avoided, so the magnetic force is also used for the propulsion. Therefore, there is no rotational component needed. In this sense, the vehicle is no longer limited by adhesion in traction and many related issues like rolling resistance, rolling contact fatigue and rail squeal noise. Non-contact linear motors are used for both traction and braking.

The linear motor is providing the propulsion in maglev and hyperloop systems. The linear motor is an electric motor which has its stator and rotor "unrolled", thus instead of producing a torque (rotation) it produces a force along its length, as shown in Figure 3.7 [46]. There are two main types of linear motors used: Linear induction motors (LIM) and Linear synchronous motors (LSM) [8,13-14]. For the linear motor, the stator with the longitudinal moving magnetic field is called primary and the rotor with induced/constant magnetic field is called secondary. In maglev systems, both the primary and the secondary can be installed either on the moving vehicle or on the fixed guideway, and vice versa. The configuration of the primary and the secondary is dependent on the installation cost of the guideway.

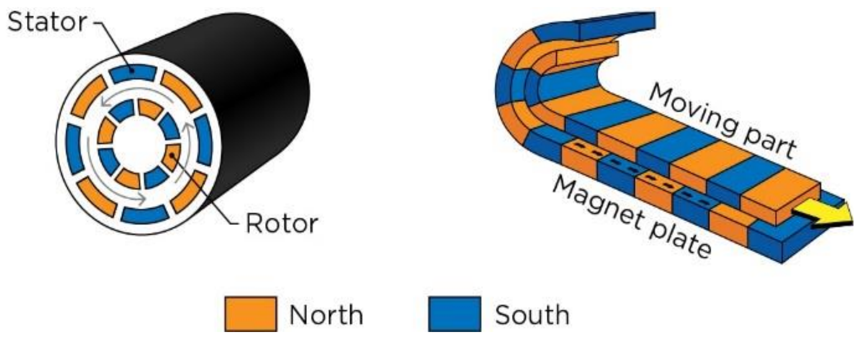
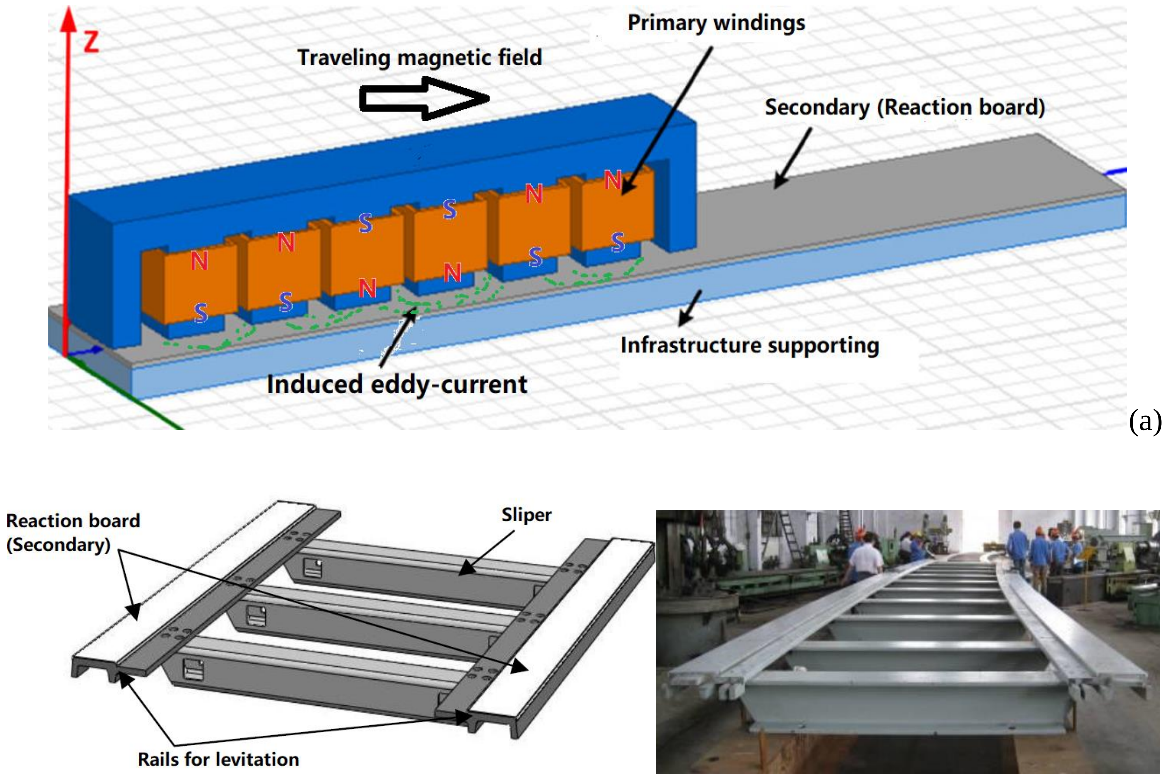


Figure 3.7: Function of linear motor.

For low-speed maglev systems, a short primary type of linear induction motor is used. In the system, the primary is installed on the vehicle and the secondary is aluminium plates (reaction board) on the guideway, as shown in Figure 3.8(a) [47]. The travelling magnetic field of the three-phase winding onboard works as the primary and can induce eddy-currents and magnetic field in the aluminium plate to propel the vehicle in the longitudinal direction. The vehicle can either be standing-still or move in the longitudinal direction and the induced eddy-current is caused by the longitudinal movement of the magnetic field. The structure of the guideway can therefore be much simplified, so the capital costs are significantly reduced, as shown in Figure 3.8(b) [48]. For this kind of system, the air gap between the primary and the reaction board should be small. Normally, to maintain a constant power supply, current collectors mounted on the vehicle are in contact with powered rails to get electricity from the infrastructure. The systems have low energy efficiency and cannot exceed speeds of around 300 km/h because of the current collectors. Realized designs are the Japanese HSST, Korean UTM and Chinese low-speed urban maglev systems.

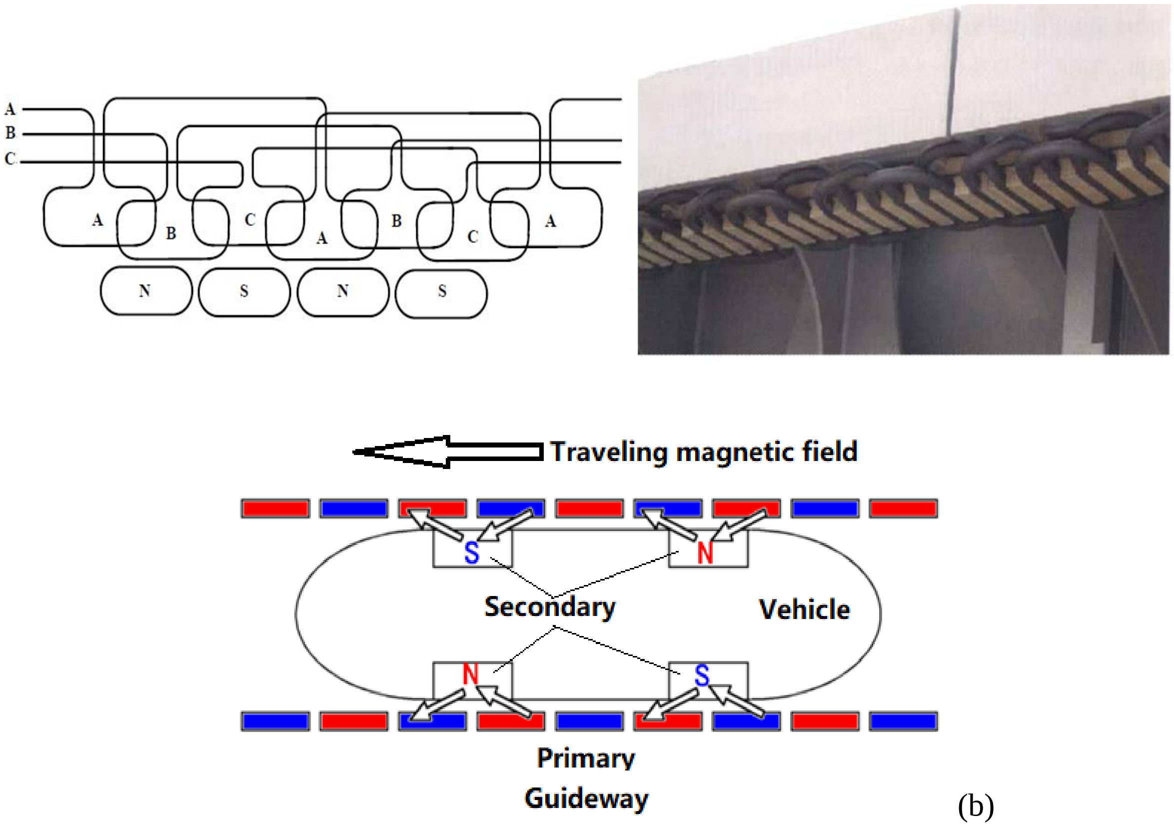


(a)

(b)

Figure 3.8: (a) Functionality of Linear induction motor of low-speed maglev systems; (b) Guideway of low-speed maglev systems in China.

For high-speed maglev systems, linear synchronous motors (LSM) are used. Different from low-speed systems, the primary is on the guideway, so the travelling magnetic field is realized by the winding on the guideway, as shown in Figure 3.9(a) [49]. Permanent magnets or electromagnets are used for the secondary on the vehicle. During operation, the secondary closely follows the travelling magnetic field, so there is no slip between the vehicle and the magnetic field of the primary, as shown in Figure 3.9(b) [50]. There is no need for a large amount of electric energy transfer to the vehicle, so current collectors can be avoided and the system is thus suitable for high-speed applications. Since the vehicle closely follows the travelling of the magnetic field of the primary on the guideway, the speed is precisely controlled by the frequency of the current in the primary. But the construction cost is much higher than for low-speed systems. Regarding the secondary, the German Transrapid maglev system uses electromagnets with iron core and the Japanese MLX maglev system uses superconducting magnets with air core.



(a)

(b)

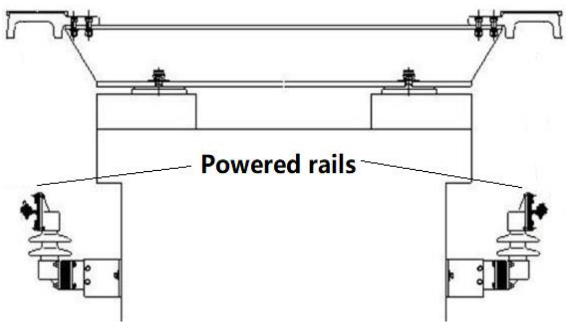
Figure 3.9: (a) Windings on the guideway; (b) Functionality of linear synchronous motor (LSM) in Japanese MLX high-speed maglev system.

3.4 Power supply

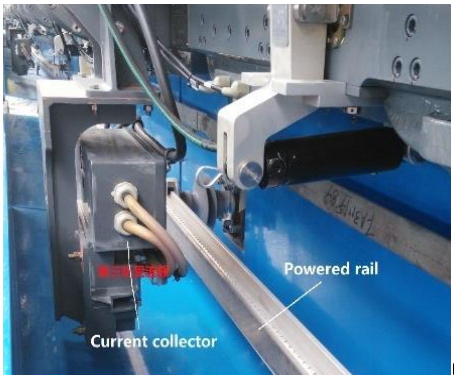
Even though all maglev trains have batteries on their vehicles, an electric power supply from the ground side is necessary for levitation, guidance, propulsion, onboard electrical equipment, battery recharging, etc. There are two methods to transfer electric energy all along the guideway: a mechanical-based current collecting system and a linear generator.

(1) Mechanical-based current collecting system and powered rail

For low-speed maglev systems, the energy transfer is very similar to metro systems, where a powered conductor (third rail) is built along the track. Since the maglev trains are lifted up and therefore lack contact with the rails, there is an additional rail for returning current. Therefore, two conducting rails are installed below the guideway, as shown in Figure 3.10(a) [40]. The current collectors are extended from the carbody to reach the two conducting rails on both sides, as shown in Figure 3.10(b) [40]. Due to limited dynamic performance, the system can only work within a low-speed range.



（a）



(b)

Figure 3.10: (a) Powered rails for low-speed maglev systems; (b) Current collector pressing against a powered rail.

(2) Linear generator

At high speeds, the maglev trains can no longer obtain power from the infrastructure by using mechanical contact. Therefore, high-speed maglev trains use non-contact methods to deliver the power to the vehicle from the infrastructure. A linear generator is therefore used, where the generator coils are onboard and induced by the change of magnetic flux from the guideway to provide electricity. The German Transrapid train employs the use of a linear generator that is integrated into the levitation electromagnets as demonstrated in Figure 3.11 [14]. The linear generator derives power from the travelling electromagnetic field when the vehicle is moving. The frequency of the generator windings is six times higher than the motor synchronous frequency. The linear generator is mechanically contact-free, which is very positive for high-speed operation. For the Japanese MLX maglev system, beside a gas turbine generator, two linear generators are mounted on the two sides.

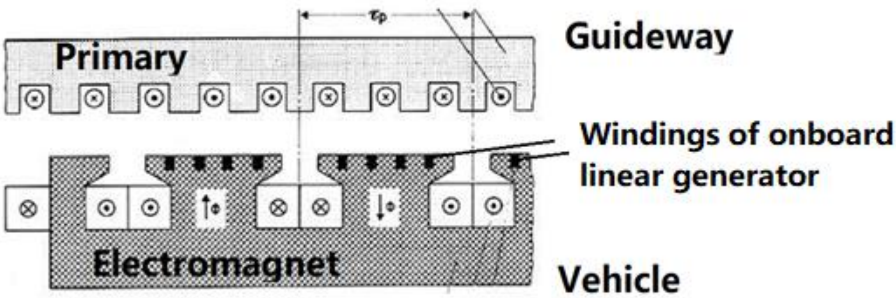


Figure 3.11: Linear generator of German Transrapid maglev system (windings of linear generator is inserted in the levitation electromagnets on the vehicle.)

3.5 Low air-pressure hyperloop concept

Maglev systems are elevated from the guideway and the moving trains have no direct contact with the guideway, so there is no rolling resistance or rotational components that rely on adhesion. However, with speed increasing, the aerodynamic resistance becomes dominating, which limits the operational speed of maglev trains. In order to achieve a low air-drag working environment for the guided vehicles, the vehicles/capsules are supposed to run inside an evacuated tube with very low air pressure, as shown in Figure 3.12 [52]. The tube-based vehicle is propelled and guided by the guideway inside the “vacuum” tube and is supposed to run at a speed near the speed of sound. The levitation, guidance, and propulsion and braking are supposed to be the same system as in maglev systems.

The moving vehicle experiences aerodynamic and magneto-dynamic forces, internal structural dynamic forces, and internal forces created by the payload as well as the vehicle subsystems with moving components including pumps and turbomachinery. The loads are transferred to the guideway and then to the tube, including gravitational forces, propulsion forces, centripetal forces during curving, and forces transferred from the control system to minimize vibrations and perturbations on the vehicle. Conceptually, there are many merits of the hyperloop over other transport systems. Compared to aircraft, tube-based vehicles are protected from atmospheric phenomena such as turbulence, microbursts, storms, lightning strikes, and collisions with flocks of birds because the tube is sealed. Compared with other ground-based vehicles, the vehicles are electrically driven in a controlled guideway with reduced air resistance, so the achievable operational speed can be very high with low energy consumption.

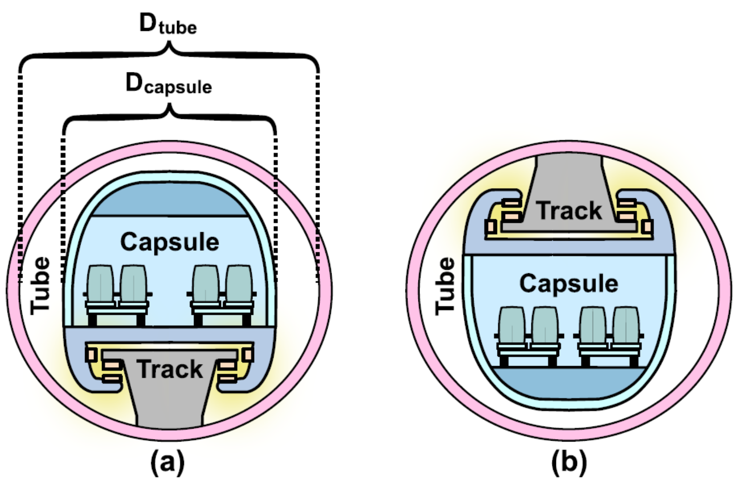


Figure 3.12: Illustration of the hyperloop transport system. (a) The vehicle floats on the guideway. (b) the vehicle hangs underneath the guideway.

The pressurized vehicle/pod is supposed to run in the evacuated tubes and boarding and alighting are handled at specially-designed terminals. The evacuated condition can be realized by pumping the air out of the tube. Vacuum pumps are used to remove the air from the sealed tube to maintain the desired atmospheric pressure and deal with gas leakage that occurs along the tube from connections. These vacuum pumps are placed along the connected tubes at discontinuous locations. However, it is very difficult to maintain a vacuum over large distances and it is impossible to achieve a true vacuum condition. Therefore, the air pressure in the sealed tube is kept at 100 Pa (about 1/1000 of the air pressure at sea level). In a reduced air-pressure environment for high-speed motion, there are still limits to vehicle speed, since the tube is not perfectly vacuumed.

For any vehicle moving through the air in a confined channel, tube or guideway, this limit is known as the Kantrowitz limit. It applies to trains, automobiles, as well as the TransPod system.

More specifically, the Kantrowitz limit describes choked flow, where a contraction in the cross-sectional area causes an increase in flow velocity according to the continuity equation. At the limit when this velocity reaches the speed of sound, the flow becomes choked, and no further increase in flow rate can be achieved, despite changes in pressure upstream or downstream. An aerodynamic shaping of the vehicle can assist the bypass of air, but the potential for choked flow in the bypass cross-section presents a fundamental limit. In general, there are three categories of aerodynamic strategies to overcome the Kantrowitz limit for a vehicle travelling in a confined tube [20]:

- Tube diameter sufficiently large to allow sufficient air bypass around the vehicle;
- Reduced air density (pressure) inside the tube environment;
- Auxiliary bypass of tube air, through the vehicle from the front to the back.

A disadvantage of large-diameter tube segments is that the tube dimensions greatly affect the infrastructure cost. The remaining two options provide solutions which do not require material costs continuously at each point along the infrastructure cross-section. Reducing the air pressure requires vacuum pumps at discrete points along with the infrastructure and it is not practical to achieve a 100% vacuum, so the vehicle still experiences a degree of air resistance. Auxiliary bypass is only required on the vehicle, which is much more convenient compared to measures on the entire infrastructure. It is possible to combine the above three strategies in a proportion that can be optimized based on energy efficiency, reliability or cost. In particular, the auxiliary bypass strategy can include a gas forcing system, such as an axial compressor on the vehicle to drive air through a preferred route at the reduced cross-sectional area. So most of the hyperloop vehicles have a compressor bypass under (or above) the passenger cabinet and the air is exhausted at a rocket-style nozzle, as shown in Figure 3.13 [20].

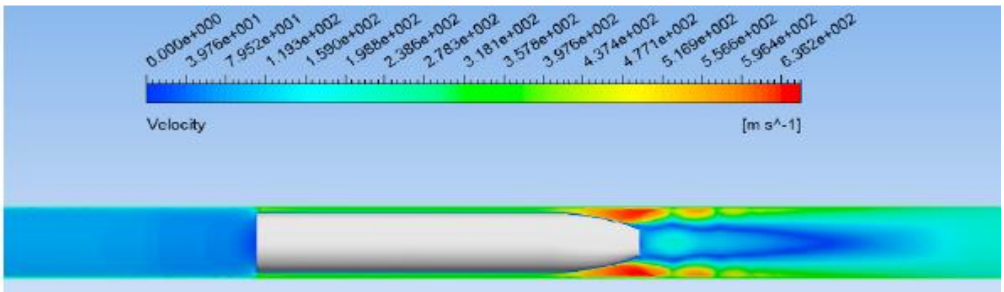


Figure 3.13: Aerodynamic simulation of the vehicle running in a low-pressurized tube.

Although there are many other technical challenges in the development of hyperloop systems, e.g. signalling system and vehicle-infrastructure dynamics, the main technical concerns are from the low air-pressure working condition. Building and maintaining low air pressure at a large scale is costly and difficult in engineering applications. The capacity of the hyperloop system is constrained by the infrastructure. Heat transfer and cooling are very poor for any onboard device in a vacuum condition. In an emergency, evacuation and rescue become very difficult in the sealed low-pressurized tube. Therefore, different pressure levels greatly determine the vehicle design and the safety concept. For example, fire and smoke behave differently in low-pressure tube environments.

3.6 Summary and comparison

There are many types of suspension systems as mentioned above. The different features of the suspension systems are summarized and compared in Table 3.1.

Table 3.1: Comparison of different types of the maglev suspension system

Characeristic	**Electro-magnetic suspension (EMS)**	**Electro-dynamic suspension (EDS)**	**Superconducting magnetic suspension (SMS)**
Type of mode	Attraction force	Repulsive force	Repulsive force
Magnets	Iron cored electromagnets	Superconducting coils	Superconductor
Cooling	Not necessary	Necessary	Necessary
Guideway gap	10-15 mm	100-150 mm	10-20 mm
Guideway components	Laminated strips	Aluminium strips	Permanent magnets
Feedback control	Necessary	Unnecessary	Unnecessary
Compatible drive system	Linear induction motor	Linear synchronous motor	Linear synchronous motor
Example	Transrapid	MLX (Japanese L0)	-

4 Applications

Maglev systems have been proposed and developed for several decades, whereas hyperloop systems are relatively new. Several maglev systems (and test systems) have been built or are under construction, while there is no commercial hyperloop system built or under construction.

Six commercial maglev systems are currently in operation around the world [53]. One is located in Japan, two in South Korea, and three in China. In Aichi, Japan, a system built for the 2005 World's Fair, the Linimo, is still in operation. It is about 9 km long, with nine station stops over that distance, and a maximum speed of about 100 km/h. The Korean Rotem Maglev runs in the city of Daejeon between the Daejeon Expo Park and the National Science Museum with a distance of 1 km. The Inchon Airport Maglev has six stations and runs from Inchŏn International Airport to the Yongyu station, 6.1 km away. The longest commercial maglev system is in Shanghai; it covers about 30 km and runs from the urban area to Pudong International Airport. The line is the first high-speed commercial maglev, operating at a maximum speed of 430 km/h. China also has two low-speed maglev systems operating at speeds of 100 km/h. The Changsha maglev connects that city's airport to a station 18.5 km away, and the S1 line of the Beijing subway system has seven stops over a distance of 9 km. Japan is building a long-distance high-speed maglev system, the Chuo Shinkansen, by 2027 that should connect Nagoya to Tokyo, a distance of 286 km, with an extension to Osaka (514 km from Tokyo) planned for 2037. The Chuo Shinkansen is planned to travel at 500 km/h. Meanwhile, there are several maglev systems and hyperloop systems built for testing. This chapter will take several representative systems as examples to summarize and compare the features of different systems in the application or testing stage.

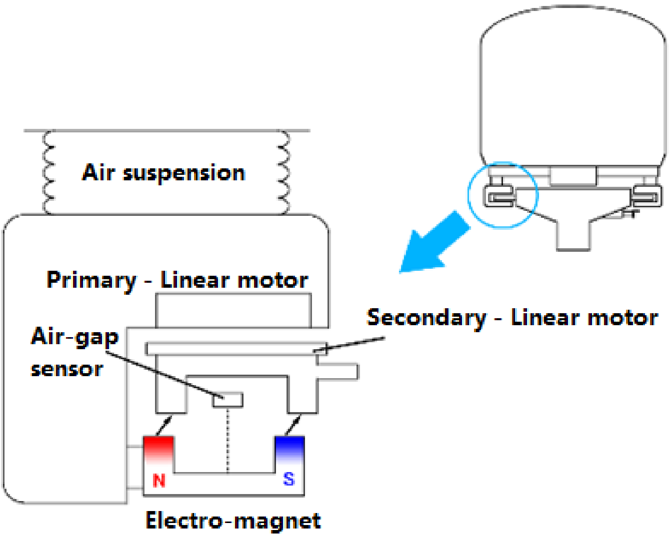
4.1 Low-speed maglev lines

There are five low-speed maglev lines in commercial operation in the world. All of them use electromagnetic suspension (EMS) technology and their maximum operational speeds are around 100 km/h. The systems are based on attractive forces generated by onboard electromagnets to levitate and guide the vehicle and a short primary type of linear induction motor onboard for propulsion and braking. This kind of system can cope with sharp curves and large gradients, and the construction cost is relatively low.

(1) Japanese High-speed surface transport (HSST) maglev system in Aichi

Japan started developing a low-speed maglev system in 1972 and the first HSST-01 maglev train was finished in 1975. The Linimo line was built in 2005 in the Aichi prefecture, Japan, to serve the Expo 2005 fair site, as shown in Figure 4.1 [54]. Now the line operates to serve the local community. The line is 8.9 km-long with the narrowest curve radius of 75 m and a maximum gradient of 6%. The vehicle is manufactured by Nippon Sharyo and has a total length of 43.3 m with three cars and the maximum acceleration and deceleration can reach 4 m/s^2. The top operational speed is 100 km/h. The daily ridership is about 16 000.

(a)



(b)

Figure 4.1: Japanese Linimo line in Aichi. (a) the vehicle and track. (b) the functionality of the suspension system.

(2) Korean low-speed maglev systems in Daejeon and Incheon

The development of low-speed maglev systems started in Korea in the mid-1980s. The first prototype train was built in 1992. For Expo 93 in Daejeon, a 1 km-long maglev track was finished for a demonstration to the public [31]. Now the line is used for transport between the Expo Park and the National Science Museum with an annual number of passengers of around one million. The train consists of two cars and the top speed is 100 km/h with a maximum acceleration of 3.6 m/s^2, as shown in Figure 4.2. Later, a 6.1 km-long urban maglev line opened in 2016 at the Incheon airport [55]. The two-car trains are made by Hyundai Rotem and run at a top speed of 110 km/h with a capacity of 230 persons per train.


(a)

(b)

Figure 4.2: Daejeon maglev line. (a) the UTM-02 train and track. (b) the interior of the train.


(a)

(b)

Figure 4.3: Incheon airport maglev line. (a) the train and track. (b) the interior of the train.

(3) Chinese urban maglev systems in Changsha and Beijing

The development of the Chinese urban maglev systems began in 1989. The first prototype was finished in Chengdu in 1994. The first commercial line started its operation in 2016 in Changsha with a system length of 18.5 km. The train consists of three cars with a top operational speed of

140 km/h and a capacity of 363 persons, as shown in Figure 4.4 [56]. The maximum daily ridership is over 12 000. The second urban maglev system in China is 9.1 km long and opened in 2017 [57]. The line is part of the Beijing Metro network. The train in operation consists of six cars with a top operational speed of 100 km/h and a capacity of 1032 persons, as shown in Figure 4.5. The maglev trains in both Changsha and Beijing are manufactured by CRRC.

(a) (b)

Figure 4.4: Changsha Maglev Express. (a) the train and track. (b) the interior of the train.

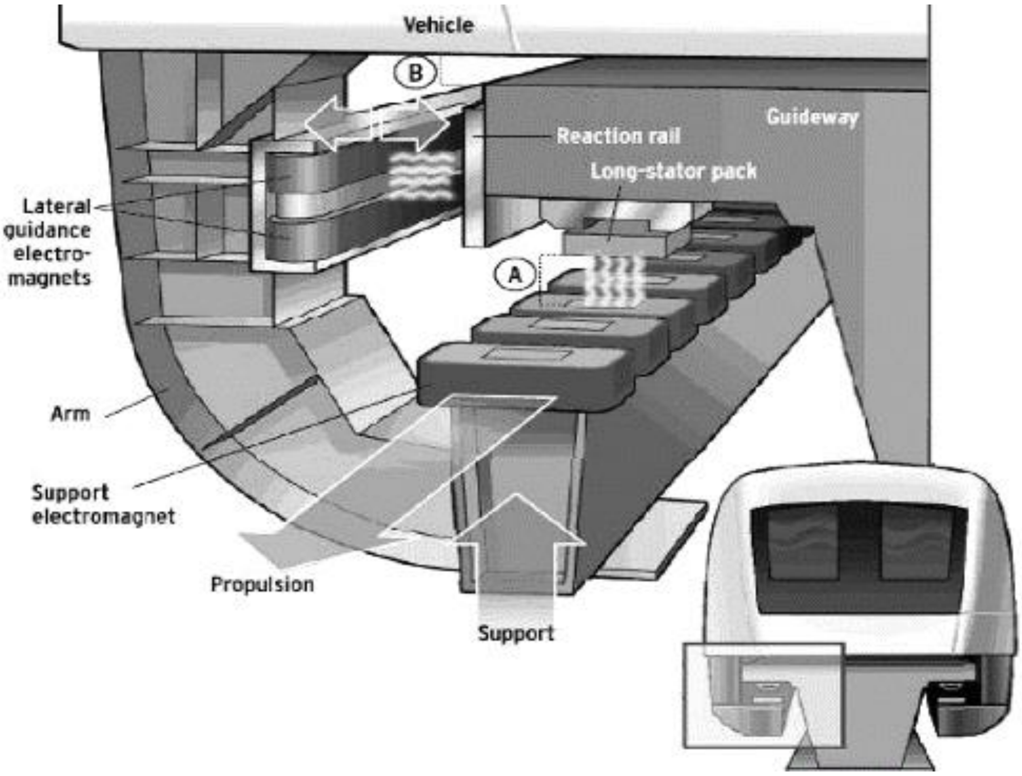
(a) (b)

Figure 4.5: Beijing Line S1. (a) the train and track. (b) the interior of the train.

4.2 German Transrapid maglev system and Shanghai maglev line

Germany started its research on high-speed maglev systems in 1969. Transrapid was a joint venture of Siemens and ThyssenKrupp. There were a series of Transrapid maglev trains developed and tested. The latest version is Transrapid 09 which is 75.8 m-long and designed for an operational speed of 500 km/h with an acceleration of 1 m/s^2, as shown in Figure 4.6 [14, 33]. The Transrapid maglev systems use electromagnetic suspension in both vertical and lateral directions for levitation and guidance, respectively. The propulsion and braking are carried out by linear synchronous motor with the primary on the guideway. Although there were several attempts to build commercial lines in Germany, no commercial maglev system is built up to now. In September 2006, a Transrapid train collided with a stand-still maintenance vehicle at 170 km/h on the test track and 23 fatalities were reported, which was the first fatal accident on any maglev system [33]. The Emsland test track closed down in 2011 and was dismantled in 2012. The last Transrapid 09 maglev train is displayed in a museum.

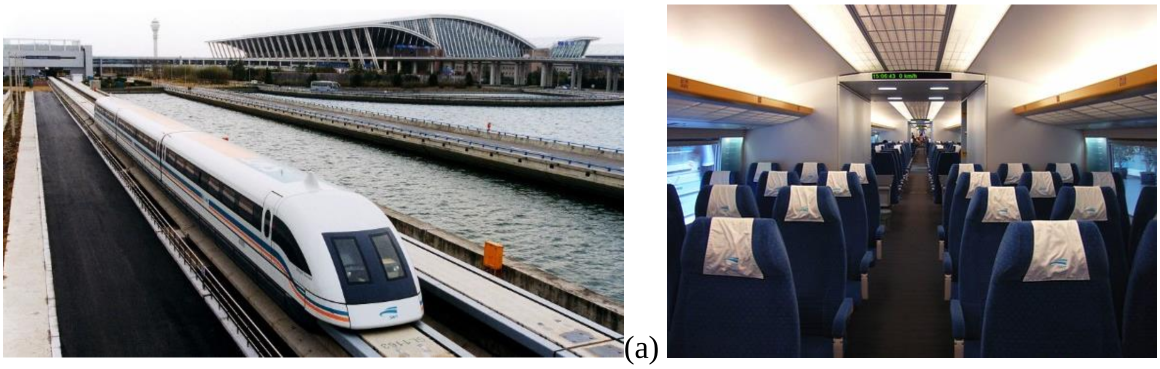

(a)



(b)

Figure 4.6: German Transrapid maglev train. (a) Transrapid 09 maglev train and track in Emsland. (b) Functionality of the suspension system.

The German Transrapid maglev system was implemented in Shanghai, China, in 2002 [58]. It was the first and still the only commercial high-speed maglev system in the world. The maglev line is 30.5 km-long connecting Shanghai Pudong International airport to the urban area. The narrowest curve radius is 4.4 km. The maglev train is based on Transrapid 08 and has a top operational speed of 430 km/h, see Figure 4.7. The train consists of four or five cars with a capacity of 959 seats. In 2003, it achieved a top speed of 501 km/h in a test run. The maximum daily ridership was 20 000. In 2006 a maglev train caught fire as it arrived at the terminal due to a flawed battery. No deaths or injuries were reported.



(a) (b)

Figure 4.7: Shanghai maglev train system. (a) the train and track. (b) the interior of the train.

4.3 Chinese high-speed maglev system

China imported the German Transrapid maglev system to Shanghai in 1999, meanwhile own research on high-speed maglev systems started in the same year. In 2013 the 10-year service contract with Transrapid expired and the Chinese partners took over all the technology and maintenance tasks. Based on the Transrapid maglev system, the first maglev prototype train, manufactured by CRRC, was unveiled in Qingdao in 2019, as shown in Figure 4.8 [59-60]. It uses electro-magnetic suspension and is designed for a top speed of 600 km/h. The train has a flexible configuration from 2 to 10 cars to meet different needs and each car can accommodate about 100 seated passengers. The prototype train was tested on a test line at Tongji University in Shanghai in 2020. It still needs some time to validate the performance of this maglev system before it can be tested at the designed speed. Although China has the largest high-speed railway network in the world, the maglev system is developed to fill the gap between the current high-speed trains (350 km/h) and aeroplanes (800 km/h) to support the national comprehensive transport network plan for 2035.


(a)

(b)


(c)

(d)

Figure 4.8: Chinese high-speed maglev train unveiled in Qingdao. (a) Train and track. (b) Interior of the train. (c) Switch for high-speed maglev train. (d) Test track at Tongji University in Shanghai.

4.4 Japanese L0 maglev train and Chuo Shinkansen

Japan started the development of high-speed maglev systems based on electro-dynamic suspension (EDS) in the 1960s and achieved a top speed of 517 km/h in 1977 on a seven km-long test line with electromagnetic technology. Since 1979 the superconducting technology is applied to the high-speed maglev trains [34]. In 1997 the Yamanashi maglev test line was finished, which is 42.8 km long and based on a commercialized standard. In 2003 a three-car MLX01 test train reached a maximum speed of 581 km/h. In 2010 the L0 high-speed maglev train was unveiled on the Yamanashi maglev test line. After a series of test runs, the seven-car L0 test train broke the previous world record and reached a top speed of 603 km/h (keeping the speed for 10.8 s) on the test track in 2016, as shown in Figure 4.9 [60-62]. The L0 series maglev train is designed for an operational speed of 500 km/h. The bogies are arranged in a Jacobs bogie configuration. The L0 series test train has a flexible configuration, which can be changed between 5 and 12 cars. Each car is 2.9 m wide, 3.1 m high, and the end cars and intermediate cars are 28 m and 24 m long, respectively. In the 12-car configuration, the train is 299 m long and has a capacity of 728 seats. The manufacturer of the L0 maglev train is a joint venture of Mitsubishi Heavy Industries, Nippon Sharyo and Hitachi Rail.


(a)

(b)

(c)

(d)

Figure 4.9: Janpanese L0 high-speed maglev. (a) Train in a test run. (b) Interior of the train. (c) Train in the U-shape track. (d) Space for mounting bogie.

The Japanese L0 high-speed maglev system uses electro-dynamic suspension (EDS) system [40]. The functionality is as shown in Figure 4.10(a). Both the suspension and guidance systems are mounted on the two sides of the guideway. There are two kinds of coils mounted on the two sides of the guideway. The levitation/guidance coils on the two sides of the guideway are magnetized by the superconducting coil when the train is passing by, so the repulsive forces between the levitation/guidance coils and the superconducting coil can elevate and guide the vehicle. Since EDS does not work at low speeds, supporting wheels and guidance wheels in the bogie are used at low speeds and in any emergency, as shown in Figure 4.10(c). Since superconductors are applied, a complex cooling system is used, as shown in Figure 4.10(d). Liquid helium is used to cool down the superconductor to achieve superconducting. Since liquid helium needs to be preserved at a very low temperature, liquid nitrogen is used to cool down the liquid helium and a special cooling system is implemented. The EDS system can maintain a large air gap (more than 100 mm) at high speeds and the superconducting coil can maintain elevation for a while without any power, so it is not sensitive to environmental disturbances, structural errors and dynamic loads, which makes it suitable and safe for high-speed operation. There is no need to have a complex active system to control the air gap. However, since the magnetic field is quite strong and the magnetic flux leakage is quite high in the EDS system, it is necessary to screen out the magnetic flux leakage in the passenger cars. In addition, only low-magnetic steel or high-strength carbon fibre can be used to build the guideway, which increases the construction cost.

The L0 series maglev train and the MLX maglev system have proven in a series of tests that they can be used for commercial operation, so the Japanese government approved the construction of the Chuo Shinkansen line in 2014. The Chuo Shinkansen is a 505 km-long maglev line for an operational speed of 500 km/h, which connects Tokyo and Nagoya and will be extended to Osaka, as shown in Figure 4.11. The first phase of the project to Nagoya is supposed to be finished by 2027 [60]. About 86% of the 286 km-long section will be in tunnels. The entire project is supposed to be finished by 2037. At that time, the travelling time between Tokyo and Osaka will be reduced from 160 min by Shinkansen high-speed railway trains to 67 min by the maglev train. The maglev station in Tokyo will be built underground about 40 m below the existing Shinkansen station in the city centre. According to the budget, the construction cost of the first phase will be 5 trillion JPY (about 400 billion SEK), i.e., 11 billion JPY/km (about 1.4 billion SEK/km).

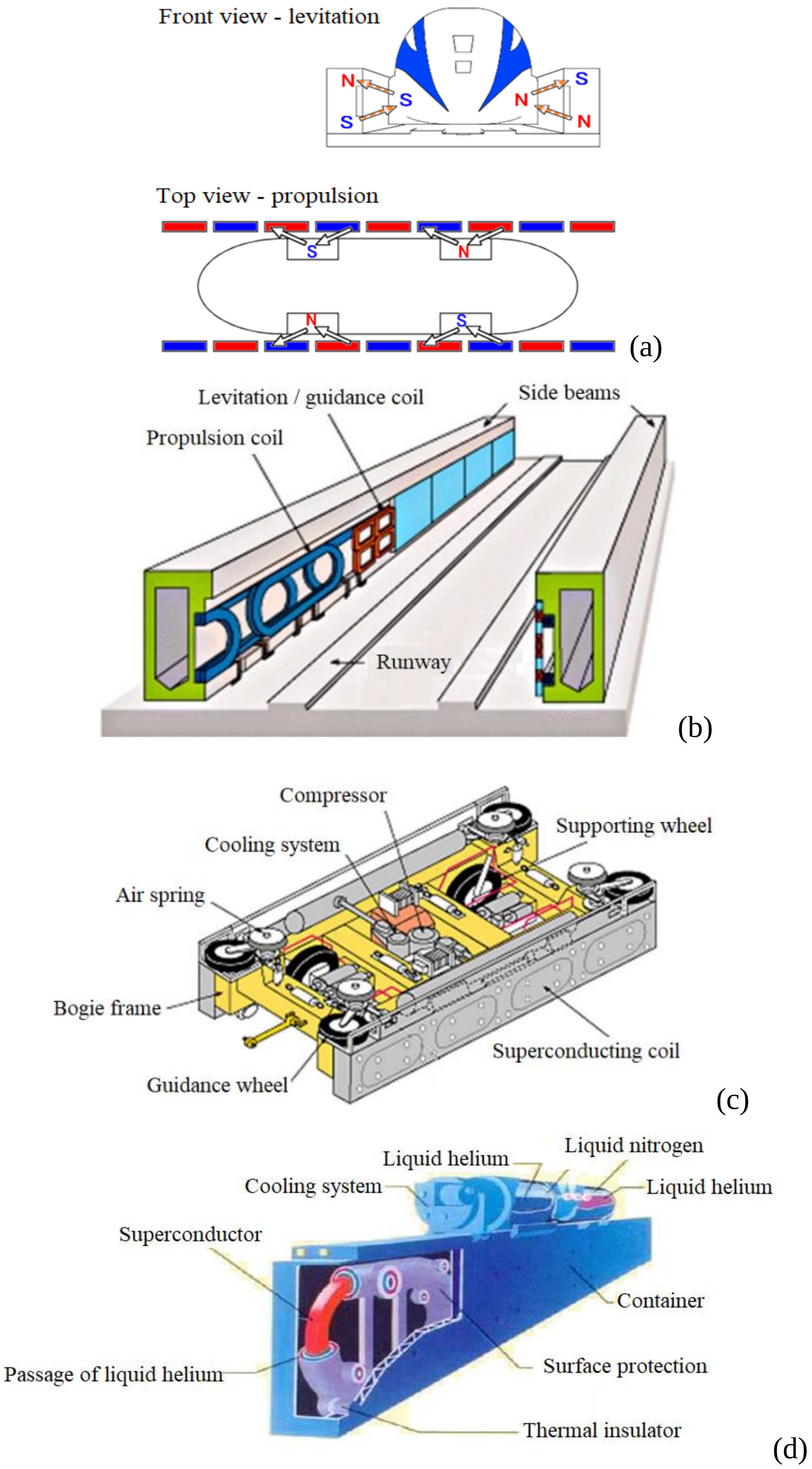


Figure 4.10: Functionality of Japanese L0 maglev train and MLX maglev system. (a) Mechanism of suspension, guidance and propulsion. (b) Structure of the guideway. (c) Structure of bogie. (d) Structure of the superconducting coil on the two sides of the bogie.

Figure 4.11: Japanese maglev system under construction.

4.5 Chinese high-temperature superconducting maglev system

Copper-based oxide superconductors were discovered in 1986 and some superconductors with critical transition temperatures higher than liquid nitrogen have been discovered. This makes it possible to use liquid nitrogen, which is rich in resources and inexpensive, as a coolant for superconductors. From 2000, a maglev suspension system based on high-temperature superconductors started to be developed in China. In 2021, a prototype vehicle and a short section of track based on this technology were unveiled at Southwest Jiaotong University in China, as shown in Figure 4.12 [41,64]. This type of maglev train is reported to be designed for a speed of 620 km/h. The superconductors made of $YBa_2Cu_3O_7$, are mounted on the bogie frames under the carbody and permanent magnets are mounted on the guideway. The linear motor is placed along the track in the middle of the guideway. The structure of this type of suspension system is relatively simple because the guideway and the superconductors can both support and guide the vehicle. There is no need to have an active system to control the air gap. The supporting force and air gap are not speed-dependant. The suspension system is much lighter than the other suspension systems.

Even though this type of suspension system has many merits compared to other systems, there are some limitations which prevent it from commercial applications today. Firstly, the supporting capacity is still low and the air gap is relatively small, which implies a small structural error and displacement need to be ensured in high-speed operation. Secondly, the permanent magnet guideway uses a large number of neodymium iron boron permanent magnet materials, which make the construction cost very high. Thirdly, the permanent magnetic material track has a strong attraction to ferromagnetic metal, so in the long term of very high-speed operation, it is necessary to keep the guideway clear from attracted metals.

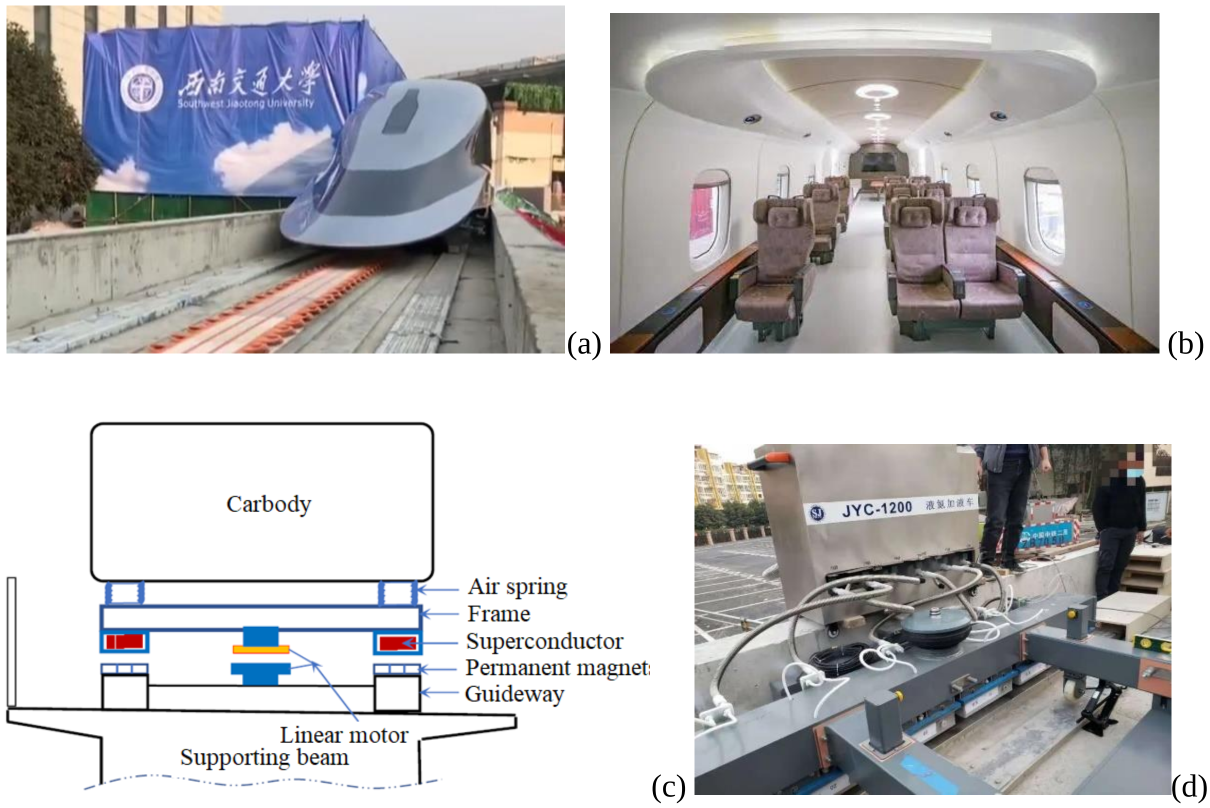


Figure 4.12: Chinese high-temperature superconducting maglev system. (a) Train on a short section of track for demonstration. (b) Interior of the train. (c) Mechanism of suspension. (d) Bogie and superconductor refilled with liquid nitrogen.

4.6 Hyperloop

As discussed above, there are some maglev systems aiming at operational speeds of 400 - 620 km/h under development or even in construction in the world, but the aerodynamic resistance and airborne noise constrain further increase of the operational speed. Therefore, hyperloop systems operating in low air-pressure sealed conditions are a possible solution for very high-speed operation. Since this concept is relatively new and a lot of relevant technologies are still in testing or under development, there are no designs or products near commercialisation. A one-km-long test track with a diameter of 1.8 m was sponsored by SpaceX and built in California in 2016 for the hyperloop pod competition. Up to now, there are some hyperloop systems developed and tested in different countries.

In the United States, the hyperloop concept was developed by Virgin Hyperloop (formerly Hyperloop Technologies, Hyperloop One and Virgin Hyperloop One). The original Hyperloop concept is proposed to use a linear electric motor to accelerate and decelerate an air-bearing levitated pod through a low-pressure tube to run at a speed of 1200 km/h. Now the hyperloop system is suspended by a magnetic system in a vacuum tube. The original company was established in 2013. In 2017 a test track was completed outside Las Vegas, which is 500 m-long with a diameter of 3.3 m, as shown in Figure 4.13 [65]. The full-scale passenger pod XP-1 reached a speed of 387 km/h without any passenger in the sealed tube in the same year. The XP-1 pod is 8.7 m long, 2.7 m wide and 2.4 m tall, as shown in Figure 4.14 [66]. In 2020 the next generation of passenger pod XP-2 was released and completed the first test with two passengers at a speed of 172 km/h, as shown in Figure 4.15 [67]. The test was conducted in a near-vacuum environment of 100 Pa. Virgin's goal is to stress test the entire system by 2025, then have commercial operations for passengers and cargo in effect by the end of the decade. The final pods would hold around 20 people, each moving directly to a destination, with more pods at busier times to cope with demand. The commercial version of the hyperloop pod has a capacity of 28 passengers and was unveiled in 2021.

 (a)  (b)

Figure 4.13: (a) 500 m-long test track in Nevada. (b) Interior of the sealed tube.

Figure 4.14: Full scale Hyperloop pod XP-1 under testing.

(a) 
(b)

Figure 4.15: Hyperloop pod XP-2 with two seats. (a) Exterior. (b) Interior.

Hyperloop Transportation Technologies, also known as HyperloopTT is another American company developing the hyperloop concept, which was founded in 2013. The system uses an electromagnetic suspension (EMS) system and is propelled by a linear induction motor. In 2019 a 320 m-long full-scale test track with a diameter of 4 m was built in Toulouse, France, as shown in Figure 4.16(a), while a prototype hyperloop pod is under development, as shown in Figure 4.16(b) [68]. The pod is 32 m-long with a tube diameter of 2.7 m and can accommodate up to 40 passengers.

(a) 

(b)

Figure 4.16: Hyperloop system of the Hyperloop Transportation Technologies. (a) Test track in Toulouse, France. (b) Prototype pad Quintero One for demonstration in Spain.

Some countries are interested in hyperloop systems. For example, based on the existing high-temperature superconducting maglev system, a demonstration system was finished at Southwest Jiaotong University in 2013 with a total length of 45 m and tube diameter of 2 m, as shown in Figure 4.17(a). A down-scale sealed test platform was built at Southwest Jiaotong University, as shown in Figure 4.17(b) [41]. A full-scale hyperloop test platform with a total length of 22 km started construction in May 2021 [69]. Hyperloop systems are also developed in Canada, the Netherlands, Korea, Spain and Poland. Figure 4.18 shows the Canadian TransPod Hyperloop system for both passengers and freight, which looks like a wingless aeroplane jet fuselage but is driven by magnetic propulsion [36].


(a)

(b)

Figure 4.17: (a) Hyperloop in China for demonstration. (b) Down-scale test platform.

Figure 4.18: Proposed Transpod-M2A hyperloop pod.

4.7 Comparision of different systems

This section compares different kinds of maglev and hyperloop systems in commercial operation or under testing/development. Compared with railway transport, these two kinds of guided transport systems are relatively new without many commercialized examples, especially for high-speed applications.

The only commercialized high-speed maglev system is, as mentioned above, the German Transrapid maglev system built in Shanghai, China. The total project cost about 9 billion CNY in 2001 (8 billion SEK). Now the top operational speed is 430 km/h and it is only kept for a short while. Figure 4.19 shows the speed profile and the power curve during a typical service run [70]. We can notice that the power demand is high during the acceleration phase. Meanwhile, a series of running tests have been performed on the maglev line. In 2003 the maglev train reached a top speed of 501 km/h and the energy usage with different top speeds was recorded. Table 4.1 [70] lists the mean energy usage of the Shanghai maglev train compared with the German ICE high-speed train and an estimation provided by Transrapid [70]. We can notice that the energy usages of both the railway and maglev systems significantly increases with running speed. At the same operational speed, the maglev systems are more energy-efficient.

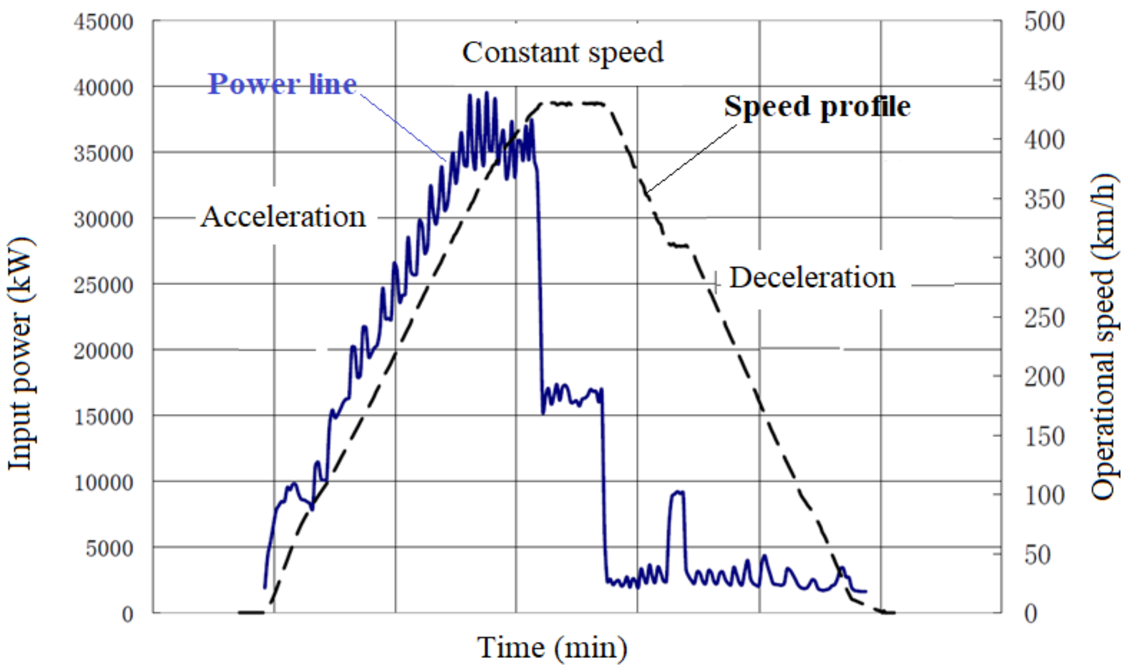


Figure 4.19: Speed profile and input power as a function of time of Shanghai maglev line.

Table 4.1: Energy usage of ICE train, Transrapid 08 and Shanghai maglev (Wh/seat/km)

Top operational speed (km/h)	200	250	300	350	400	430	450	470
German ICE train -8 cars	41	57	74	-	-	-	-	-
Tranrapid 08 - Estimation	33	37	45	-	63	-	-	-
Shanghai maglev train - measurement	36	38	46	56	65	70	74	79

Guided transport systems are developed at different stages in the world. High-speed railway systems have been developed and widely implemented for several decades and reach today top operational speeds of 350 km/h in commercial service. The maglev systems have also been developed for several decades but have just started implementation for commercial application in the past years with top operational speeds ranging from 500 km/h and 600 km/h. The hyperloop system was proposed only recently, and many technologies and components are either under conceptual development or in the testing phase. Even though the hyperloop system is designed for an operational speed above 1000 km/h, the maximum speed reached up to now in a test is 387 km/h without any passenger onboard (172 km/h with two passengers onboard), which is far below the high-speed railway (French TGV 574 km/h in test), the maglev system (Japanese L0 maglev train 603 km/h in test) and the passenger aeroplane (Concorde 2330 km/h in test). That means that there is still a long way to go to reach the targeted operational speed (1200 km/h).

Table 4.2 [70-75] compares the performances of different transport systems which are used or being developed nowadays. The hourly transport capacity is a very important key performance indicator, which can be calculated based on the capacity of each train and the hourly travel frequency of trains. We can see that both high-speed railway trains and maglev systems have high capacity and can transport more than 5 000 passengers per hour, whereas the aeroplane can only transport about 3000 passengers and 840 passengers are the limit for the hyperloop system. The high-speed railway system even has the capability to transport more than 15 000 passengers per hour in one direction. Regarding energy efficiency, both high-speed railway trains and maglev systems are much more efficient than aeroplanes. No full-scale hyperloop system has been built for high-speed operation, so there is no relevant data for comparison today. All guided transport systems have a relatively high construction cost and cannot compete with the aeroplanes in that respect, since there is no infrastructure between airports needed for the air traffic. Even though maglev and hyperloop systems are suitable for long-distance travelling, the construction of the infrastructure cost increases linearly with the length of the track, in contrast to air traffic.

Different guided transport systems have different features. Although railway systems have been developed and used for quite a long time, maglev and hyperloop systems have many advantages over traditional railway trains and many new technologies on maglev and hyperloop systems are been developed or tested. Meanwhile, for high-speed railway systems, there are also many innovations on its way, e.g. increasing the maximum operational speed from 350 km/h to 360 km/h or even 400 km/h on existing dedicated high-speed lines, increasing the capacity with new train designs, improving energy efficiency with a lightweight structure and efficient traction system, and reducing noise emission.

Table 4.2: Comparison of different transport systems

Type of system		High-speed train	German Transrapid	Japanese MLX	Virgin Hyperloop	Aeroplane
Operational condition		Beijing-Shanghai CRH400AF-B	Shanghai Maglev line	L0 series on Chuo Shinkansen	XP-2 on DevLoop test track	Boeing 777-200ER
Top operational speed	km/h	350	430	500	1 200*	890
Max. speed in test	km/h	486 (2010)	501 (2003)	603 (2016)	387 (2017 without passenger), 172 (2020, with two passengers)	950
Capacity	Seat	1283	446	1 000 (now 728)	28* (now 2)	223
Frequency	Train/ h	12	12	10	30*	15
Hourly transport capacity	Seat/h	15 396	5 352	10 000	840*	3 345
Train length	m	440	153	392 (now 296)	25* (now 5.5)	64
Weight	ton/seat	0.74	0.7	0.42	(Now 1.25)	1.26
Energy supply		Electricity	Electricity	Electricity	Electricity	Kerosene
Energy usage	Wh/seat/km	46	70	74	-	150
Technology maturity		Commercial operation	Commercial operation	Commerical line under construction	Technology under development	Commercial operation
Construction cost	million SEK/km	200 (2011 in China)	260 (2001 in China)	1 400 (2014 in Japan)	680* (estimation in USA in 2016)	0 (in between airports)

* Theoretical estimation based on the original design

5 Concluding remarks

Maglev transport systems have been proposed and developed for about 100 years but there are still very few commercial lines operating today. Although the hyperloop systems have a short history, there are many systems proposed and tested in recent years. These systems have different features compared to the well-developed rail transport. This work provides an overview of the maglev and hyperloop systems in the world with respect to their development history, core technologies and applications.

The maglev and hyperloop systems are suitable for high-speed operation above 400 km/h. They do not have any contact between the moving vehicle and the guideway, so the operational speed is not limited by the adhesion at the rail-wheel interface to provide support, guidance and traction/braking force. There is neither any rolling resistance nor any rotational component on the moving vehicle, which makes the high-speed operation safer and more energy-efficient for the same top speed. Compared with rail transport, many technical problems can be avoided, e.g., rail-wheel wear and fatigue, rolling noise, low adhesion etc. At present, there is no commercial railway train designed for an operational speed of 400 km/h, but the maglev systems have a mature and reliable technology for commercial service at top speeds between 400 km/h and 500 km/h. Even higher speed can be achieved by hyperloop systems in the future.

Maglev and hyperloop systems have a very high infrastructure capital cost. They are designed for medium and long-distance travel. Existing rail transport can already cope with an operational speed of 350 km/h. Although maglev and hyperloop systems can help to increase the top operational speed by more than 20% compared with the traditional high-speed railway system, it would probably cost much more than 20% of the increase in capital cost, which makes the new technologies (maglev or hyperloop system) less attractive. For a country like Sweden, with most of the population living in a relatively small part of the country, the time-saving for most of the travellers using a maglev system would be quite marginal, but the capital cost and the probable ticket price would increase significantly. Also, there is no interoperability between the new system and the old railway systems, so not all travellers can get benefits from the new system. Maglev and hyperloop systems do not have a higher capacity than the railway system, so there would not be any big improvement in the performance. In addition, the hyperloop system still needs a lot of time and monetary effort in system development before it can be used. It is not clear when hyperloop technology can reach the level for commercial service.

In summary, the maglev and hyperloop systems have some distinguished advantages in high-speed operation, which can be supplementary to the widely-developed railway systems. However, before the capital cost in construction and a high number of passengers can be proved to be economically feasible for commercial operation, traditional rail transport should still be the priority. Building up competencies concerning these new technologies can help to judge future developments and to participate in joint future studies, e.g. in the framework of the upcoming Europe’s Rail Joint Undertaking.

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